1 Osnove teorije

1.1 BCS teorija pri končni temperaturi

BCS model, kjer je $\xi_{\mathbf{k},\sigma} = \epsilon_{\mathbf{k},\sigma} - E_F$ in $k := (\mathbf{k}, \uparrow)$, $-k := (-\mathbf{k}, \downarrow)$:

$$H = \sum_k \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - V \sum_{k,k'} c_k^\dagger c_{-k}^\dagger c_{-k'} c_{k'} \tag{1}$$

Naredimo približek povprečnega polja, definiramo $b_k = c_k c_{-k}$ in $b_k = \langle b_k \rangle + \delta b_k = F_k + \delta b_k$, sledi (do $\mathcal{O}(\delta b_k^2)$)

$$b_k^\dagger b_{k'} \simeq F_k^* F_{k'} + F_k^* \delta b_{k'} + F_{k'} \delta b_k^\dagger = F_k^* b_{k'} + F_{k'} b_k^\dagger - F_k^* F_{k'} = \langle b_k^\dagger \rangle b_{k'} + b_k^\dagger \langle b_{k'} \rangle - \langle b_k^\dagger \rangle \langle b_{k'} \rangle \tag{2}$$

Iz česar sledi:

$$H = \sum_k \begin{pmatrix} c_k^\dagger & c_{-k} \end{pmatrix} \begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta_k^* & -\xi_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} + \frac{|\Delta|^2}{V} + \sum_k \xi_k \tag{3}$$

Ob definiciji $\Delta_k = \sum_{k'} V(k-k') \langle c_{k'} c_{-k'} \rangle$, vzamemo še lokalno interakcijo, zato je $V(q) \simeq V$ in $\Delta_k \simeq \Delta$. Naredimo Bogoljubov rotacijo, privzamemo $u_k, v_k \in \mathbb{R}$ in simetrijo okoli E_F: $u_k = u_{-k}$ ter $v_k = v_{-k}$. Iz CCR sledi $u_k^2 + v_k^2 = 1$. Kvazidelci so definirani z rotacijo

$$\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix} \tag{4}$$

nov Hamiltonjan pa postane

$$H = \sum_k \begin{pmatrix} \alpha_k^\dagger & \alpha_{-k} \end{pmatrix} \begin{pmatrix} \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) - \xi_k v_k^2 & -2\xi_k u_k v_k + \Delta_k u_k^2 - \Delta_k^* v_k^2 \\ -2\xi_k u_k v_k - \Delta_k v_k^2 + \Delta_k^* u_k^2 & \xi_k v_k^2 - \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix}, \tag{5}$$

kjer zahtevamo, da so izvendiagonalni členi enaki 0. Iz CCR zahteve lahko vzamemo parameterizacijo (tu sem vzel kompleksno nastavimo, da je $\Delta = \Delta^* e^{i2\phi_k}$):

$$v_k = \sin\left(\frac{\theta_k}{2}\right) e^{i\phi_k} \tag{6}$$

$$u_k = \cos\left(\frac{\theta_k}{2}\right) \tag{7}$$

$$u_k v_k = \cos\left(\frac{\theta_k}{2}\right) \sin\left(\frac{\theta_k}{2}\right) = \frac{1}{2} \sin\theta_k \tag{8}$$

$$u_k^2 - v_k^2 = \cos^2\left(\frac{\theta_k}{2}\right) - \sin^2\left(\frac{\theta_k}{2}\right) = \cos\theta_k \tag{9}$$

skupaj dobimo (od tu naprej $\Delta := |\Delta|$):

$$\xi_k \sin(\theta_k) = \Delta \cos(\theta_k) \tag{10}$$

kar fiksira

$$\tan(\theta_k) = \frac{|\Delta_k|}{\xi_k}. \tag{11}$$

Tako iz trikotnika preberemo

$$\sin\theta_k = \frac{|\Delta_k|}{E_k} = \frac{|\Delta_k|}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \tag{12}$$

$$\cos\theta_k = \frac{\xi_k}{E_k} = \frac{\xi_k}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \tag{13}$$

$$\tag{14}$$

kjer je $E_k = \pm\sqrt{\xi_k^2 + |\Delta_k|^2}$ energija kvazidelcev:

$$H = \sum_k E_k \alpha_k^\dagger \alpha_k - E_k \alpha_{-k}^\dagger \alpha_{-k} + \xi_k + \frac{|\Delta|^2}{V} \,. \tag{15}$$

V termičnem stanju so diagonalni delci porazdeljeni po

$$\langle \alpha_k^\dagger \alpha_k \rangle = \frac{1}{1+\exp(E_k/T)} = \frac{1}{2} - \frac{1}{2}\tanh\left(\frac{E_k}{2T}\right) \,. \tag{16}$$

Uporabni sta zvezi $f(-E/T) = 1 - f(E/T)$ in $1 - 2f(E/T) = \tanh(E/(2T))$. V definiciji ureditvenega parametra je

$$\langle c_k c_{-k} \rangle = \left\langle \left(u_k \alpha_k - v_k \alpha_{-k}^\dagger\right)\left(u_k \alpha_{-k} + v_k \alpha_k^\dagger\right) \right\rangle = u_k v_k \left(1 - \langle \alpha_{-k}^\dagger \alpha_{-k}\rangle - \langle \alpha_k^\dagger \alpha_k \rangle\right) = \frac{1}{2}\sin\theta_k \tanh\left(\frac{\beta E_k}{2}\right) \tag{17}$$

iz česar sledi enačba za režo pri $T > 0$ (seštevamo po k okoli E_F):

$$\Delta = \frac{V}{2}{\sum_k}' \sin\theta_k \tanh\left(\frac{E_k}{2T}\right) = \frac{V\Delta}{2}\sum_k \frac{\tanh\left(\frac{1}{2T}\sqrt{\Delta^2+\xi_k^2}\right)}{\sqrt{\Delta^2+\xi_k^2}} \tag{18}$$

1.2 Minimalni model ekscitonskega izolatorja pri končni temperaturi

Pri izolatorju za minimalni model vzamemo dve orbitali (a, b) model tesne vezi v vsaki posebej in lokalno Coulombovo interakcijo na mestu j med elektronskima gostotama v posameznih pasovih

$$H = H_0 + H_V = \sum_{i,j}\left(\epsilon_a^{(0)}\delta_{ij} + t_{ij}^{(a)}\right) c_{a,i}^\dagger c_{a,j} + \sum_{i,j}\left(\epsilon_b^{(0)}\delta_{ij} + t_{ij}^{(b)}\right) c_{b,i}^\dagger c_{b,j} + V\sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \tag{19}$$

$$= \sum_k \varepsilon_{a,k} c_{a,k}^\dagger c_{a,k} + \sum_k \varepsilon_{b,k} c_{b,k}^\dagger c_{b,k} + V\sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \tag{20}$$

najprej diagonaliziramo vsako orbitalo posebej, kar da gole disperzije (vezavni energiji nastavimo na $\pm G/2$) (delamo v dveh dimenzijah in mrežna konstanta $a_0 = 1$):

$$\varepsilon_{a(b),\mathbf{k}} = \pm\frac{G}{2} - 2t_{a(b)}(\cos(k_x) + \cos(k_y)) \,, \tag{21}$$

v tem primeru je najmanjša razdalja med pasovoma $G_0 = G - 4(t_a - t_b)$ oziroma $G_0 = G - 8t_a$, če je $t_b = -t_a$. Za interakcijo spet naredimo približek povprečnega polja:

$$c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \approx \langle c_{a,j}^\dagger c_{a,j}\rangle c_{b,j}^\dagger c_{b,j} + \langle c_{b,j}^\dagger c_{b,j}\rangle c_{a,j}^\dagger c_{a,j} - \langle c_{a,j}^\dagger c_{b,j}\rangle c_{b,j}^\dagger c_{a,j} - \langle c_{b,j}^\dagger c_{a,j}\rangle c_{a,j}^\dagger c_{b,j} + \text{konst.} \tag{22}$$

od tod sledi

$$H_V^{\text{MF}} = Vn_b\sum_j c_{a,j}^\dagger c_{a,j} + Vn_a\sum_j c_{b,j}^\dagger c_{b,j} - V\phi\sum_j c_{b,j}^\dagger c_{a,j} - V\phi^*\sum_j c_{a,j}^\dagger c_{b,j} + NV|\phi|^2 \tag{23}$$

kjer smo definirali:

$$n_a = \langle c_{a,j}^\dagger c_{a,j}\rangle = \frac{1}{N}\sum_k \langle c_{a,k}^\dagger c_{a,k}\rangle \tag{24}$$

$$n_b = \langle c_{b,j}^\dagger c_{b,j}\rangle = \frac{1}{N}\sum_k \langle c_{b,k}^\dagger c_{b,k}\rangle \tag{25}$$

$$\phi = \langle c_{a,j}^\dagger c_{b,j}\rangle = \frac{1}{N}\sum_k \langle c_{a,k}^\dagger c_{b,k}\rangle \tag{26}$$

$$\Delta = -\frac{V}{N}\sum_k \langle c_{a,k}^\dagger c_{b,k}\rangle, \quad \Delta^* = -\frac{V}{N}\sum_k \langle c_{b,k}^\dagger c_{a,k}\rangle \tag{27}$$

$$\Delta_k = \frac{1}{N}\sum_{k'} V(k-k')\langle a^\dagger_{k'} b_{k'}\rangle, \quad V(q) = \int e^{-i\mathbf{q}\cdot\mathbf{r}} V(\mathbf{r})\ \mathrm{d}\mathbf{r} \tag{28}$$

Uporabimo FT še na celotnem Hamiltonjanu

$$c^\dagger_{a,j} = \frac{1}{\sqrt{N}}\sum_k e^{-i\mathbf{k}\cdot\mathbf{r}_j} c^\dagger_{a,k} \tag{29}$$

$$H^{\mathrm{MF}} = \sum_k \left([\varepsilon_{a,k} + Vn_b] c^\dagger_{a,k} c_{a,k} + [\varepsilon_{b,k} + Vn_a] c^\dagger_{b,k} c_{b,k} + \Delta^* c^\dagger_{a,k} c_{b,k} + \Delta c^\dagger_{b,k} c_{a,k} \right) + \left(\frac{|\Delta|^2}{V} N - VNn_a n_b \right) \tag{30}$$

oziroma:

$$H^{\mathrm{MF}} = \sum_k \begin{pmatrix} c^\dagger_{a,k} & c^\dagger_{b,k} \end{pmatrix} \begin{pmatrix} \varepsilon^H_{a,k} & \Delta^* \\ \Delta & \varepsilon^H_{b,k} \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} + \text{konst.}\,, \tag{31}$$

kjer definiramo Hartree energiji $\varepsilon^H_{a(b),k} = \varepsilon_{a(b),k} + Vn_{b(a)}$. Transformacija v diagonalno / Bogoljubov bazo je podobna kot pri BCS, le da ne meša kreatorjev in anihilatorjev / ohranja število delcev:

$$\begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} \tag{32}$$

od tod po enakem postopku kot prej dobimo energiji kvazidelcev

$$E_{\alpha(\beta)}(k) = \frac{\varepsilon^H_{a,k} + \varepsilon^H_{b,k}}{2} \pm \sqrt{\left(\frac{\varepsilon^H_{a,k} - \varepsilon^H_{b,k}}{2} \right)^2 + |\Delta|^2} := \eta_k \pm \sqrt{\xi_k^2 + \Delta^2} \tag{33}$$

Za bookkeeping:

$$\left(\frac{\varepsilon^H_a - \varepsilon^H_b}{2} \right) \sin\theta_k + \Delta\cos\theta_k = 0 \implies \tan\theta_k = \frac{-2\Delta}{\varepsilon^H_a - \varepsilon^H_b} = \frac{-\Delta}{\xi_k} \tag{34}$$

$$u_k v_k = -\frac{\sin\theta_k}{2}\,. \tag{35}$$

Za samousklajeno enačbo zdaj potrebujemo

$$\langle c^\dagger_{a,k} c_{b,k}\rangle = \langle (u_k\alpha^\dagger_k + v_k\beta^\dagger_k)(-v_k\alpha_k + u_k\beta_k)\rangle = u_k v_k \left(\langle \beta^\dagger_k \beta_k\rangle - \langle \alpha^\dagger_k \alpha_k\rangle \right) = -\frac{1}{2}\sin\theta_k (f(E_\beta) - f(E_\alpha)) \tag{36}$$

od tod

$$\Delta = -\frac{V}{N}\sum_k \langle c^\dagger_{a,k} c_{b,k}\rangle = \frac{V}{N}\sum_k \frac{\Delta}{2\sqrt{\xi_k^2 + |\Delta|^2}} (f(E_\beta) - f(E_\alpha)) \tag{37}$$

Opomba, lahko definiramo tudi $n = n_a + n_b = \text{konst.} = 1$ in $n_a - n_b =: m$ od tod sledi

$$H^{\mathrm{MF}} = \sum_k \left(\left[\varepsilon_{a,k} + \frac{V}{2}n - \frac{V}{2}m \right] c^\dagger_{a,k} c_{a,k} + \left[\varepsilon_{b,k} + \frac{V}{2}n + \frac{V}{2}m \right] c^\dagger_{b,k} c_{b,k} + \Delta^* c^\dagger_{a,k} c_{b,k} + \Delta c^\dagger_{b,k} c_{a,k} \right) \tag{38}$$

od koder je v 2D Hartree gap enak:

$$G^H = G - 4(t_a - t_b) - Vm\,. \tag{39}$$

Note: v praksi je $m < 0$ V dveh dimenzijah se Hartree pasova srečata na

$$\mu_B = \frac{V}{2}n - \frac{t_a + t_b}{t_a - t_b}\frac{G - Vm}{2}\,. \tag{40}$$

Zamik je enak $V/2$, ko sta osnovna pasova simetrična in je zasedenost polovična. V primeru simetričnih pasov sta disperziji kvazidelcev

$$E_{\alpha(\beta)}(k) - V/2 = \pm\sqrt{\left(\frac{\epsilon_{a,k} - \epsilon_{b,k} - Vm}{2}\right)^2 + |\Delta|^2}\,. \tag{41}$$

Velja še

$$n_a = \frac{1}{N}\sum_k \left(u_k^2 f(E_\alpha) + v_k^2 f(E_\beta)\right) \tag{42}$$

$$n_b = \frac{1}{N}\sum_k \left(v_k^2 f(E_\alpha) + u_k^2 f(E_\beta)\right) \tag{43}$$

$$u_k^2 = \frac{1}{2}\left(1 + \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}}\right) \tag{44}$$

$$v_k^2 = \frac{1}{2}\left(1 - \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}}\right) \tag{45}$$

$$m = n_a - n_b = \frac{1}{N}\sum_{\mathbf{k}} \frac{\xi_{\mathbf{k}}}{\sqrt{\xi_{\mathbf{k}}^2 + \Delta^2}}(f(E_\alpha) - f(E_\beta)) \tag{46}$$

1.3 Numerika/psevdokoda

1.4 Odprti sistemi in modeliranje kopeli

Born-Markov master enačbe in Redfield: ločimo kopel in sistem,

$$\rho_S(t) = \mathrm{Tr}_B(\rho(t)) \tag{47}$$

Hamiltonjan razcepimo na $H = H_0 + H_I$, kjer je $H_0 = H_S \otimes \mathbb{I}_B + \mathbb{I}_S \otimes H_B$ razdenjen na podsistema, za interakcijski del pa privzamemo obliko $H_I = \sum_\alpha A_\alpha \otimes B_\alpha$. V interakcijski sliki je operator za evolucijo

$$U_0(t, t_0) = e^{-iH_0(t-t_0)} \tag{48}$$

$$\rho_I(t) = U_0^\dagger(t,0)\rho(t)U_0(t,0) = e^{iH_0t}\rho(t)e^{-iH_0t} \tag{49}$$

$$H_I(t) = U_0^\dagger(t,0)H_I U_0(t,0) = e^{iH_0t}H_I e^{-iH_0t} \tag{50}$$

Von Neumannova enačba je

$$\frac{d}{dt}\rho_I(t) = -i[H_I(t), \rho_I(t)] \tag{51}$$

$$\rho_I(t) = \rho_I(0) - i\int_0^t dt'\,[H_I(t'), \rho_I(t')] \tag{52}$$

vstavimo nazaj v komutator in se znebimo prostosnih stopenj kopeli:

$$\frac{d}{dt}\rho_S(t) = -\int_0^t ds\,\mathrm{Tr}_B\left([H_I(t), [H_I(t-s), \rho_S(t)\otimes\rho_B]]\right) \tag{53}$$

V interakcijskem členu so operatorji A_α napisani kot lastni operatorji H_S z energijami prehodov $\omega = E_b - E_a$

$$A_\alpha(\omega) = \sum_{E_b - E_a = \omega} |a\rangle\langle a|A_\alpha|b\rangle\langle b| \tag{54}$$

To zadošča relacijam $[H_S, A_\alpha(\omega)] = -\omega A_\alpha(\omega), \quad [H_S, A_\alpha^\dagger(\omega)] = \omega A_\alpha^\dagger(\omega)$ in
$e^{iH_S t} A_\alpha(\omega) e^{-iH_S t} = e^{-i\omega t} A_\alpha(\omega), \quad e^{iH_S t} A_\alpha^\dagger(\omega) e^{-iH_S t} = e^{i\omega t} A_\alpha^\dagger(\omega)$. V interakcijski sliki potem velja

$$H_I(t) = \sum_{\alpha,\omega} e^{-i\omega t} A_\alpha(\omega) \otimes B_\alpha(t) = e^{iH_B t} B_\alpha e^{-iH_B t} \tag{55}$$

evolucija sledi kot

$$\dot{\rho}_S = \sum_{\omega,\omega'} \sum_{\alpha,\beta} e^{i(\omega'-\omega)t} \Gamma_{\alpha\beta}(\omega) \left(A_\alpha(\omega) \rho_S(t) A_\beta^\dagger(\omega') - A_\beta^\dagger(\omega') A_\alpha(\omega) \rho_S(t) \right) + \text{h.c.} \tag{56}$$

kjer so prehodne matrike med stanji

$$\Gamma_{\alpha\beta}(\omega) = \int_0^\infty ds\, e^{i\omega s}\, \mathrm{Tr}_B \left(B_\alpha^\dagger(t) B_\beta(t-s) \rho_B \right) = \frac{1}{2}\gamma_{\alpha\beta}(\omega) + i S_{\alpha\beta}(\omega) \tag{57}$$

Evolucija pa ima strukturo

$$\frac{d\rho_S}{dt} = -i[H_{LS}, \rho_S(t)] + \mathcal{D}(\rho_S(t)) \tag{58}$$

z disipatorjem

$$\mathcal{D}(\rho_S) = \sum_{\omega} \sum_{\alpha,\beta} \gamma_{\alpha\beta}(\omega) \left(A_\beta(\omega) \rho_S A_\alpha^\dagger(\omega) - \frac{1}{2} \{ A_\alpha^\dagger(\omega) A_\beta(\omega), \rho_S \} \right) \tag{59}$$

oz. v splošnem

$$\mathcal{D}(\rho_S) = \sum_k \gamma_k \left(L_k \rho_S L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho_S \} \right) \tag{60}$$

V stacionarnem stanju velja

$$\mathcal{L}\rho_{\text{ness}} = 0\,. \tag{61}$$

Notes za vektorizacijo in superoperatorje:

$$\mathcal{L} u_\mu^R = \lambda_\mu u_\mu^R \tag{62}$$

$$\mathrm{vec}(ABC) = (A \otimes C^T)\mathrm{vec}(B) \tag{63}$$

$$\mathrm{Tr}(A^\dagger B) = \mathrm{vec}(A)^\dagger \mathrm{vec}(B) \tag{64}$$

$$\frac{d}{dt}|\rho\rangle = \left(-iH \otimes \mathbb{I} + i\mathbb{I} \otimes H^T + \sum_k L_k \otimes L_k^* - \frac{1}{2} \sum_k (L_k^\dagger L_k \otimes \mathbb{I} + \mathbb{I} \otimes L_k^T L_k^*) \right) |\rho\rangle \tag{65}$$

$$\langle \hat{O} \rangle(t) = \langle \mathbb{I} | \hat{O} \otimes \mathbb{I} | \rho(t) \rangle \tag{66}$$

Šibke sklopitve in perturbacijski pristopi

$$\mathcal{L} = -i[H, \cdot] + \epsilon \mathcal{D}[\cdot], \qquad \rho = \rho_0 + \epsilon \rho_1 + \epsilon^2 \rho_2 + \ldots \tag{67}$$

napišemo gostotno matriko sistema v lastni bazi, siagonalno, s klasičnimi populacijami

$$\rho_0 = \sum_n p_n |n\rangle\langle n|, \quad H|n\rangle = E_n |n\rangle \tag{68}$$

kjer velja v ničtem redu (by construction)

$$\mathcal{L}\rho_0 = -i[H, \rho_0] = 0 \tag{69}$$

V prvem redu so enačbe za NESS

$$-i[H, \rho_1] + \mathcal{D}(\rho_0) = 0 \tag{70}$$

kjer je $\mathcal{D}(\rho_0) = \sum_\mu \left(L_\mu \rho_0 L_\mu^\dagger - \frac{1}{2}\{L_\mu^\dagger L_\mu, \rho_0\} \right)$ in od tod enačba za prehode med populacijami

$$\langle n|\mathcal{D}(\rho_0)|n\rangle = \sum_\mu \sum_m p_m |\langle n|L_\mu|m\rangle|^2 - \sum_\mu \sum_m p_n |\langle m|L_\mu|n\rangle|^2 \tag{71}$$

kot klasična populacijska dinamika z $R_{nm} = \sum_\mu |\langle n|L_\mu|m\rangle|^2$

$$\dot{p}_n = \epsilon \sum_m (R_{nm} p_m - R_{mn} p_n) \tag{72}$$

Izvendiagonali elementi se razvijejo hitro na $\mathcal{O}(1)$, medtem ko se populacije razvijajo počasi na GGE manifoldu z $\mathcal{O}(\epsilon^{-1})$. Za ohranitev normalizacije mora veljati $\sum_n \dot{p}_n = 0$.

Integrabilni sistemi imajo makrsoskopsko število ohranjenih nabojev $[H_0, C_i] = 0$, pod unitarno časovno evolucijo se razvijejo v netermično stanje

$$\rho_{GGE} = \frac{1}{Z} e^{-\sum_i \lambda_i C_i} \tag{73}$$

$$\langle \psi_0 | C_i | \psi_0 \rangle = \mathrm{Tr}(C_i \rho_{\mathrm{GGE}}) \tag{74}$$

fiksira λ_i. UNitarna dinamika prijcira na GGE manifold, šibka sklopitev premika po GGE manifoldu:

$$\rho_0(t) = \rho_{\boldsymbol{\lambda}(t)} = \frac{1}{Z(t)} \exp\left(-\sum_i \lambda_i(t) C_i \right) \tag{75}$$

EOM za ohranjene naboje je

$$\langle \dot{C}_i \rangle_{\lambda(\infty)} = \mathrm{Tr}[C_i \mathcal{D}\rho] \simeq \epsilon\, \mathrm{Tr}[C_i \mathcal{D}\rho_0] = 0 \quad \forall \lambda_i \tag{76}$$

Definiramo projektor na GGE manifold:

$$PX = -\sum_{ij} \frac{\partial \rho}{\partial \lambda_i} (\chi^{-1})_{ij} \mathrm{Tr}[C_j X] \tag{77}$$

ki da dinamiko vzdolž manifolda:

$$\frac{\partial \rho_{\boldsymbol{\lambda}}}{\partial t} = \sum_i \frac{\partial \rho_{\boldsymbol{\lambda}}}{\partial \lambda_i} \dot{\lambda}_i = \epsilon P \mathcal{D}(\rho_{\boldsymbol{\lambda}}) = -\epsilon \sum_{ij} \frac{\partial \rho_{\boldsymbol{\lambda}}}{\partial \lambda_i} (\chi^{-1})_{ij} \mathrm{Tr}\left[C_j \mathcal{D}(\rho_{\boldsymbol{\lambda}})\right] \tag{78}$$

od koder sledi evolucija za koeficiente/potenciale λ_i:

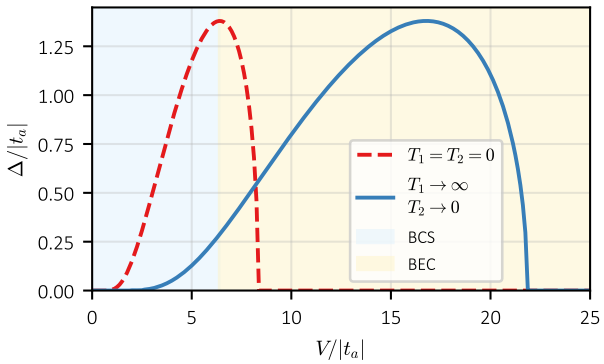
$$\dot{\lambda}_i = -\epsilon \sum_j (\chi^{-1})_{ij} \mathrm{Tr}[C_j \mathcal{D}\rho_{\lambda(t)}] \tag{79}$$

s susceptibilnostmi $\chi_{ij} = \langle C_i C_j \rangle - \langle C_i \rangle \langle C_j \rangle = -\mathrm{Tr}\left[C_i \frac{\partial \rho}{\partial \lambda_j} \right]$.

2 Ekscitonski izolatorji in sklopitev s kopelmi

$$\mathrm{Tr}\left[f \frac{1}{2}\{\rho, A\} \right] = \mathrm{Tr}\left[\frac{1}{2} f\rho A + \frac{1}{2} f A \rho \right] = \frac{1}{2} \mathrm{Tr}\left[Af\rho + fA\rho \right] \tag{80}$$

$$= \frac{1}{2} \mathrm{Tr}\left[(Af + fA)\rho \right] = \frac{1}{2} \langle Af + fA \rangle \tag{81}$$



Slika 1 | Gap pri $T_1 \to \infty$ in $T_2 \to 0$.

2.1 Osnovni modeli kopeli

2.2 Limita neskončne temperature

2.3 Fononske sklopitve

Najprej pogledamo, kako modificiran (Peierls tip) hopping integral in modificiran orbital energy vplivata na sklopitev. Model spremenimo: orbitali znotraj o.c. sta na enakem položaju in se deformirata vedno v isto smer, torej $\mathbf{u}_{a,i} = \mathbf{u}_{b,i} = \mathbf{u}_i$. Coulombski potencial se zaradi deformacije ne spremeni veliko. Hamiltonka za nedeformiran kristal ostane enaka, višje člene pa obravnavamo perturbativno:

$$H[\{\mathbf{u}\}] = \underbrace{\sum_{i,\gamma,\gamma'} \varepsilon_{\gamma\gamma',i}[\{\mathbf{u}\}]\, c^\dagger_{\gamma,i} c_{\gamma',i}}_{\text{on-site in CEF}} + \underbrace{\sum_{\langle i,j\rangle,\gamma} t_\gamma(|\mathbf{r}_{ij}|)\, c^\dagger_{\gamma,i} c_{\gamma,j}}_{\text{hopping, med oribitalami}} + \underbrace{V \sum_j n_{a,j} n_{b,j}}_{\text{Coulomb}} \tag{82}$$

Razdalje so:

$$\mathbf{r}_j = \mathbf{R}_j + \mathbf{u}_j\,, \tag{83}$$

oziroma razdalje med paroma (sosedi):

$$\mathbf{r}_{ij} = \boldsymbol{\delta} + \mathbf{u}_j - \mathbf{u}_i \;\to |\mathbf{r}_{ij}| \simeq |\boldsymbol{\delta}| + \boldsymbol{\delta}\cdot(\mathbf{u}_j - \mathbf{u}_i) + \mathcal{O}(u^2) \tag{84}$$

Predpostavimo da na hopping vpliva samo sprememba razdalje:

$$t_\gamma(|\mathbf{r}_{ij}|) \simeq t_\gamma(|\boldsymbol{\delta}|) + \boldsymbol{\delta}\cdot(\mathbf{u}_j - \mathbf{u}_i)\frac{\mathrm{d}t_\gamma(x)}{\mathrm{d}x}\bigg|_{x=|\boldsymbol{\delta}|=1} \tag{85}$$

Deformacija vpliva na vezavno energijo preko CEF, razvijemo preko deformacijskega tenzorja ($\varepsilon_{\mu\nu} = \frac{1}{2}(\partial_\mu u_\nu + \partial_\nu u_\mu)$ oziroma diskretno: $\varepsilon_{\mu\nu,j} = \frac{1}{4}(u^\nu_{j+\mu} - u^\nu_{j-\mu} + u^\mu_{j+\nu} - u^\mu_{j-\nu})$ zgornji indeks je koordinata vektorja). Če je eno elektronska Hamiltonka na mestu j (el. - ion) (po μ, ν se sešteva po Einsteinu):

$$h_j = h_0 + \delta h_j + O(\varepsilon^2), \quad \delta h_j = \frac{\partial U_{\text{el-ion}}}{\partial \varepsilon_{\mu\nu,j}} \varepsilon_{\mu\nu,j},\ (\mu,\nu) = (x,y) \tag{86}$$

in je v nedeformiranem stanju:

$$\varepsilon^{(0)}_\gamma = \langle \phi_\gamma | h_0 | \phi_\gamma \rangle \tag{87}$$

potem so popravki (lahko mešajo orbitali)

$$\delta\varepsilon_{\gamma\gamma'} = \left\langle \phi_\gamma \left| \frac{\partial U_{\text{el-ion}}}{\partial \varepsilon_{\mu\nu,j}} \right| \phi_{\gamma'} \right\rangle \varepsilon_{\mu\nu,j} = D^{\mu\nu}_{\gamma\gamma'} \varepsilon_{\mu\nu,j} \tag{88}$$

Sklopitvena Hamiltonka v realni bazi je tako ($\sum_{\langle ij\rangle} \to \sum_i \sum_{\delta=\{x,y\}}$)

$$H_{ep}^{(1)} = \sum_{i,\gamma,\gamma'} D_{\gamma\gamma'}^{\mu\nu} \varepsilon_{\mu\nu,i} c_{\gamma,i}^\dagger c_{\gamma',i} + \sum_{i,\delta,\gamma} t'_\gamma \boldsymbol{\delta} \cdot (\mathbf{u}_{i+\delta} - \mathbf{u}_i) \left(c_{\gamma,i}^\dagger c_{\gamma,i+\delta} + \text{h.c.} \right) \tag{89}$$

(note):

$$\sum_{\boldsymbol{\delta}} \boldsymbol{\delta} \cdot (\mathbf{u}_{i+\delta} - \mathbf{u}_i) = \hat{x} \cdot (\mathbf{u}_{i+x} - \mathbf{u}_i) - \hat{x} \cdot (\mathbf{u}_{i-x} - \mathbf{u}_i) + \hat{y} \cdot (\mathbf{u}_{i+y} - \mathbf{u}_i) - \hat{y} \cdot (\mathbf{u}_{i-y} - \mathbf{u}_i) \approx 2(\varepsilon_{xx} + \varepsilon_{yy}) \tag{90}$$

Zdaj kvantiziramo odmike in prepišemo v k-prostor, specificno za 2D square lattice:

$$\mathbf{u}_j = \frac{1}{\sqrt{N}} \sum_{\mathbf{q},\lambda} \frac{1}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \boldsymbol{\epsilon}_{\mathbf{q}\lambda} e^{i\mathbf{q}\cdot\mathbf{R}_j} \left(b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger \right) \tag{91}$$

v prvem členu so sami členi oblike (od defomracijskega tenzorja, diskretna verzija)

$$u_{j+\delta}^\mu - u_{j-\delta}^\mu = \frac{2i}{\sqrt{N}} \sum_{\mathbf{q},\lambda} \frac{\epsilon_{\mathbf{q}\lambda}^\mu}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \sin(q_\delta)\, e^{i\mathbf{q}\cdot\mathbf{R}_j} \left(b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger \right). \tag{92}$$

v drugem pa členi ($\delta = \mu = \{x, y\}$, glej note):

$$\frac{t'_\gamma}{\sqrt{N}} \sum_{\mathbf{q},\mathbf{k}',\mathbf{k}} \frac{\epsilon_{\mathbf{q}\lambda}^\mu}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \left(b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger \right) \left(e^{iq_\mu} - 1 \right) \left(e^{ik_\mu} + e^{-ik'_\mu} \right) c_{\gamma,\mathbf{k}'}^\dagger c_{\gamma,\mathbf{k}} \sum_j \frac{1}{N} e^{i(\mathbf{q}+\mathbf{k}-\mathbf{k}')\cdot\mathbf{R}_j} = \tag{93}$$

$$= \frac{t'_\gamma}{\sqrt{N}} \sum_{\mathbf{k},\mathbf{q}} \frac{\epsilon_{\mathbf{q}\lambda}^\mu}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \left(b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger \right) \left(e^{iq_\mu} - 1 \right) \left(e^{ik_\mu} + e^{-i(k_\mu+q_\mu)} \right) c_{\gamma,\mathbf{k}+\mathbf{q}}^\dagger c_{\gamma,\mathbf{k}} = \tag{94}$$

$$= \frac{2it'_\gamma}{\sqrt{N}} \sum_{\mathbf{k},\mathbf{q}} \frac{\epsilon_{\mathbf{q}\lambda}^\mu}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \left(\sin(k_\mu + q_\mu) - \sin(k_\mu) \right) c_{\gamma,\mathbf{k}+\mathbf{q}}^\dagger c_{\gamma,\mathbf{k}} \left(b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger \right) \tag{95}$$

Hamiltonka je na koncu oblike (intraband = $\gamma = \gamma'$):

$$H_{ep} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k},\mathbf{q},\lambda} \psi_{\mathbf{k}+\mathbf{q}}^\dagger \begin{pmatrix} g_{aa}^\lambda(\mathbf{k},\mathbf{q}) & g_{ab}^\lambda(\mathbf{q}) \\ g_{ba}^\lambda(\mathbf{q}) & g_{bb}^\lambda(\mathbf{k},\mathbf{q}) \end{pmatrix} \psi_{\mathbf{k}} \left(b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger \right) \tag{96}$$

$$g^\lambda(\mathbf{k},\mathbf{q}) = \frac{i}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \begin{pmatrix} 2t'_a T^\lambda(\mathbf{k},\mathbf{q}) + F_{aa}^\lambda(\mathbf{q}) & F_{ab}^\lambda(\mathbf{q}) \\ F_{ba}^\lambda(\mathbf{q}) & 2t'_b T^\lambda(\mathbf{k},\mathbf{q}) + F_{bb}^\lambda(\mathbf{q}) \end{pmatrix}. \tag{97}$$

kjer so:

$$T^\lambda(\mathbf{k},\mathbf{q}) = \sum_{\mu=x,y} \epsilon_{\mathbf{q}\lambda}^\mu \left[\sin(k_\mu + q_\mu) - \sin(k_\mu) \right] \tag{98}$$

$$F_{\gamma\gamma'}^\lambda(\mathbf{q}) = D_{\gamma\gamma'}^{xx} \epsilon_{\mathbf{q}\lambda}^x \sin(q_x) + D_{\gamma\gamma'}^{yy} \epsilon_{\mathbf{q}\lambda}^y \sin(q_y) + D_{\gamma\gamma'}^{xy} \left[\epsilon_{\mathbf{q}\lambda}^y \sin(q_x) + \epsilon_{\mathbf{q}\lambda}^x \sin(q_y) \right]. \tag{99}$$

interband členi so neničelni, glede na izbrane orbitale (glej (88)) in izbran CF potencial, odvisno od simetrije. Ker smo izbrali $\mathbf{u}_{a,i} = \mathbf{u}_{b,i}$ se ti elementi sklapljajo samo z akustičnimi fononi, ce znotraj o.c. ni prostostnih stopenj(??).

2.4 Optični fononi

Rabimo dodatne ligande / dodatne ione znotraj celice. Še vedno obdržimo $\mathbf{u}_{a,j} = \mathbf{u}_{b,j} = \mathbf{u}_j$, okolica pa se deformira neodvisno. Odmik liganda ℓ v celici j razvijemo po normalnih nacinih

$$\mathbf{v}_{j,\ell} = \sum_s Q_{s,j}\, \mathbf{e}_{s,\ell}, \qquad \sum_\ell \mathbf{e}_{s,\ell} \cdot \mathbf{e}_{s',\ell} = \delta_{ss'}, \tag{100}$$

kjer je s indeks nacina, $Q_{s,j}$ skalarna amplituda. Razvijemo enoelektronsko hamiltonko

$$h_j = h_0 + \sum_s \left.\frac{\partial U_{\text{el-ion}}}{\partial Q_{s,j}}\right|_{Q=0} Q_{s,j} + O(Q^2), \qquad \Lambda^s_{\gamma\gamma'} = \left\langle \phi_\gamma \left| \frac{\partial U_{\text{el-ion}}}{\partial Q_{s,j}} \right| \phi_{\gamma'} \right\rangle_0 . \tag{101}$$

$$H_{ep}^{\text{opt}} = \sum_{j,s} \sum_{\gamma,\gamma'} \Lambda^s_{\gamma\gamma'} Q_{s,j}\, c^\dagger_{\gamma,j} c_{\gamma',j}. \tag{102}$$

Kvantiziramo koordinate okoliških ionov:

$$Q_{s,j} = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}} \frac{1}{\sqrt{2M_s\omega_{\mathbf{q}s}}} e^{i\mathbf{q}\cdot\mathbf{R}_j} \left(b_{\mathbf{q}s} + b^\dagger_{-\mathbf{q}s} \right), \tag{103}$$

od tod

$$H_{ep}^{\text{opt}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k},\mathbf{q},s} \psi^\dagger_{\mathbf{k}+\mathbf{q}}\, g^s(\mathbf{q})\, \psi_{\mathbf{k}} \left(b_{\mathbf{q}s} + b^\dagger_{-\mathbf{q}s} \right), \qquad g^s(\mathbf{q}) = \frac{\Lambda^s}{\sqrt{2M_s\omega_{\mathbf{q}s}}}. \tag{104}$$

Kar da končne sklopitve pri $\mathbf{q} \to 0$ ($\Omega_s \equiv \omega_{\mathbf{q}\to 0,s}$):

$$g^s(\mathbf{q} \to 0) = \frac{\Lambda^s}{\sqrt{2M_s\Omega_s}}. \tag{105}$$

2.5 Fotonske sklopitve v resonatorju

Začnemo z minimalnim modelom za elektrone:

$$H_e = H_e^{(0)} + H_V = \sum_k \varepsilon_{a,k} c^\dagger_{a,k} c_{a,k} + \sum_k \varepsilon_{b,k} c^\dagger_{b,k} c_{b,k} + V \sum_j c^\dagger_{a,j} c_{a,j} c^\dagger_{b,j} c_{b,j} \tag{106}$$

in privzamemo 2D plast, kjer sta dve orbitali na o.c., ki hibridizirata samo preko Coulombske interakcije. Disperzije za model neinteragirajoč sistem so v tem primeru enake (21). Interakcija pa je v prvem redu oblike:

$$H_{e\gamma}^{(1)} = -\int \mathrm{d}\mathbf{r}\, \mathbf{j}_e \cdot \mathbf{A} = -\int \mathrm{d}\mathbf{r} \sum_{\mathbf{q}} \mathbf{A}(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{r}} \mathbf{j}_e(\mathbf{r}) = -\sum_{\mathbf{q}} \mathbf{A}(\mathbf{q}) \cdot \mathbf{j}_e(-\mathbf{q}), \tag{107}$$

kjer je

$$\mathbf{A}(\mathbf{q}) = \sum_\lambda \frac{1}{\sqrt{2\mathcal{V}\epsilon_0\omega_{\mathbf{q},\lambda}}} \boldsymbol{\epsilon}_{\mathbf{q},\lambda} \left(a_{\mathbf{q},\lambda} + a^\dagger_{-\mathbf{q},\lambda} \right) \tag{108}$$

V t.b. modelu dobimo tok iz kontinuitetne in Heisenbergove enačbe:

$$\frac{\partial \rho(\mathbf{r})}{\partial t} + \nabla \cdot \mathbf{j} = 0 \;\to\; i\mathbf{q} \cdot \mathbf{j}(\mathbf{q}) = -\frac{\partial \rho(\mathbf{q})}{\partial t} \tag{109}$$

(+ je od IBP) in

$$\mathbf{q} \cdot \mathbf{j}_e(\mathbf{q}) = i\frac{\partial \rho(\mathbf{q})}{\partial t} = -[H_e, \rho(\mathbf{q})] \tag{110}$$

Elektronska gostota je

$$\rho(\mathbf{r}) = -e \cdot \psi^\dagger(\mathbf{r})\psi(\mathbf{r}) \tag{111}$$

operatorje polja razvijemo po lokaliziranih / Wannierovih orbitalah

$$\psi(\mathbf{r}) = \sum_{i,\gamma=\{a,b\}} \phi_\gamma(\mathbf{r} - \mathbf{R}_i) c_{\gamma i}\,. \tag{112}$$

Transformiramo s FT, uvedemo novo spremenljivko $\mathbf{r} = \mathbf{R}_i + \mathbf{x}$ in razvijemo do dipolov $e^{-i\mathbf{q}\cdot\mathbf{x}} \simeq 1 - i\mathbf{q}\cdot\mathbf{x}$ (delamo se da se orbitale ne prekrivajo izven o.c., torej $\sum_{ij} \to \sum_i \delta_{ij}$):

$$\rho(\mathbf{q}) = -e \sum_{i,j} \sum_{\gamma,\gamma'} \int \mathrm{d}\mathbf{r}\; e^{-i\mathbf{q}\cdot\mathbf{r}} \phi^*_\gamma(\mathbf{r} - \mathbf{R}_i) \phi_{\gamma'}(\mathbf{r} - \mathbf{R}_j) c^\dagger_{\gamma i} c_{\gamma' j} = -e \sum_{i,\gamma,\gamma'} e^{-i\mathbf{q}\cdot\mathbf{R}_i} \langle \phi_\gamma | e^{-i\mathbf{q}\cdot\mathbf{x}} | \phi_{\gamma'} \rangle c^\dagger_{\gamma i} c_{\gamma' i} \tag{113}$$

Zdaj pa recimo da je dipolni moment samo med različnima orbitalama v o.c. in $n_{\gamma i} = \langle \phi_\gamma | \phi_\gamma \rangle c^\dagger_{\gamma,i} c_{\gamma,i}$:

$$\rho(\mathbf{q}) \simeq -e \sum_i e^{-i\mathbf{q}\cdot\mathbf{R}_i} \left(n_{ai} + n_{bi} - i\mathbf{q}\cdot\mathbf{d} \left(c^\dagger_{a,i} c_{b,i} + c^\dagger_{b,i} c_{a,i} \right) \right) \tag{114}$$

zdaj uporabimo zvezo

$$\sum_i e^{i\mathbf{q}\cdot\mathbf{R}_i} c^\dagger_{ai} c_{bi} = \sum_{\mathbf{k}} c^\dagger_{a,\mathbf{k}+\mathbf{q}} c_{b\mathbf{k}} \tag{115}$$

z negativnim argumentom je potem elektronska gostota do dipolnega člena (upoštevajoč $\mathbf{d} = \langle \phi_a(\mathbf{x}) | \mathbf{x} | \phi_b(\mathbf{x}) \rangle$):

$$\rho(-\mathbf{q}) = -e \sum_{\mathbf{k}} \begin{pmatrix} c^\dagger_{a,\mathbf{k}+\mathbf{q}} & c^\dagger_{b,\mathbf{k}+\mathbf{q}} \end{pmatrix} \left(\mathbb{1} + i\,\mathbf{q}\cdot\mathbf{d} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) \begin{pmatrix} c_{a\mathbf{k}} \\ c_{b\mathbf{k}} \end{pmatrix} \tag{116}$$

Zdaj poračunamo komutator, da dobimo tok. Delam z vektorji $\psi_k = \begin{pmatrix} c_{a,\mathbf{k}} & c_{b,\mathbf{k}} \end{pmatrix}^\mathsf{T}$:

$$[H^{(0)}_e, \rho(\mathbf{q})] = -e \sum_{\mathbf{k},\mathbf{k}'} \left[\psi^\dagger_{\mathbf{k}} \begin{pmatrix} \varepsilon_{a,\mathbf{k}} & 0 \\ 0 & \varepsilon_{b,\mathbf{k}} \end{pmatrix} \psi_{\mathbf{k}}, \psi^\dagger_{\mathbf{k}'-\mathbf{q}} \begin{pmatrix} 1 & -i\mathbf{q}\cdot\mathbf{d} \\ -i\mathbf{q}\cdot\mathbf{d} & 1 \end{pmatrix} \psi_{\mathbf{k}'} \right] \tag{117}$$

uporabimo zvezo:

$$\left[\psi^\dagger_{\mathbf{k}} A \psi_{\mathbf{k}}, \psi^\dagger_{\mathbf{k}'-\mathbf{q}} B \psi_{\mathbf{k}'} \right] = \psi^\dagger_{\mathbf{k}'-\mathbf{q}} \Big(A(\mathbf{k}'-\mathbf{q}) B(\mathbf{k}') - B(\mathbf{k}'-\mathbf{q}) A(\mathbf{k}') \Big) \psi_{\mathbf{k}'}, \delta_{\mathbf{k},\mathbf{k}'-\mathbf{q}} \tag{118}$$

$$= \psi^\dagger_{\mathbf{k}} \Big(A(\mathbf{k}) B(\mathbf{k}+\mathbf{q}) - B(\mathbf{k}) A(\mathbf{k}+\mathbf{q}) \Big) \psi_{\mathbf{k}+\mathbf{q}} \tag{119}$$

od tod

$$\mathbf{q}\cdot\mathbf{j}_e(\mathbf{q}) = -[H^{(0)}_e, \rho(\mathbf{q})] = -e \sum_{\mathbf{k}} \psi^\dagger_{\mathbf{k}} \begin{pmatrix} \varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}} & i(\mathbf{q}\cdot\mathbf{d}) \left(\varepsilon_{a,\mathbf{k}} - \varepsilon_{b,\mathbf{k}+\mathbf{q}} \right) \\ i(\mathbf{q}\cdot\mathbf{d}) \left(\varepsilon_{b,\mathbf{k}} - \varepsilon_{a,\mathbf{k}+\mathbf{q}} \right) & \varepsilon_{b,\mathbf{k}+\mathbf{q}} - \varepsilon_{b,\mathbf{k}} \end{pmatrix} \psi_{\mathbf{k}+\mathbf{q}} \tag{120}$$

da izrazimo tok, naredimo približek, da prispevajo le majhni $\mathbf{q} \to 0$, zato:

$$\varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}} \simeq \mathbf{q} \cdot \nabla_{\mathbf{k}} \varepsilon_{\mathbf{k}+\mathbf{q}/2}\,, \tag{121}$$

kar drži točno za proste elektrone in približno za tight binding disperzije. In general ce ne expandamo:

$$\mathbf{j}_e(\mathbf{q}) = -\frac{e\mathbf{q}}{\|\mathbf{q}\|^2} \sum_{\mathbf{k}} \psi^\dagger_{\mathbf{k}} M(\mathbf{k},\mathbf{q}) \psi_{\mathbf{k}+\mathbf{q}} \;+\; \mathbf{q} \times \mathbf{C}(\mathbf{q}) \tag{122}$$

kjer so $\mathbf{C}(\mathbf{q})$ arbitrarna vektorska polja (tokovi brez divergence, magnetizacijski tokovi). Ce pa naredimo ta priblizek pa dobimo za tok v splošnem, indekse zamaknemo za $-\mathbf{q}$:

$$\mathbf{j}_e(\mathbf{q}) \simeq -e \sum_{\mathbf{k}} \psi^\dagger_{\mathbf{k}-\mathbf{q}} \begin{pmatrix} \nabla_{\mathbf{k}} \varepsilon_{a,\mathbf{k}-\mathbf{q}/2} & i\mathbf{d}(\varepsilon_{a,\mathbf{k}-\mathbf{q}} - \epsilon_{b\mathbf{k}}) \\ i\mathbf{d}(\varepsilon_{b,\mathbf{k}-\mathbf{q}} - \varepsilon_{a\mathbf{k}}) & \nabla_{\mathbf{k}} \varepsilon_{b,\mathbf{k}-\mathbf{q}/2} \end{pmatrix} \psi_{\mathbf{k}}\,. \tag{123}$$

Od tod splošna Hamiltonka v prvem redu:

$$H^{(1)}_{e\gamma} = -\sum_{\mathbf{q}} \mathbf{j}_e(-\mathbf{q}) \cdot \mathbf{A}(\mathbf{q}) = \tag{124}$$

$$= \sum_{\mathbf{k},\mathbf{q},\lambda} \frac{e}{\sqrt{2\mathcal{V}\varepsilon_0 \omega_{\mathbf{q},\lambda}}} \psi^\dagger_{\mathbf{k}+\mathbf{q}} \underbrace{\begin{pmatrix} \boldsymbol{\epsilon}_{\mathbf{q}\lambda} \cdot \mathbf{v}_a\left(\mathbf{k}+\frac{\mathbf{q}}{2}\right) & i\left(\boldsymbol{\epsilon}_{\mathbf{q}\lambda}\cdot\mathbf{d}\right)\left(\varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{b,\mathbf{k}}\right) \\ i\left(\boldsymbol{\epsilon}_{\mathbf{q}\lambda}\cdot\mathbf{d}\right)\left(\varepsilon_{b,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}}\right) & \boldsymbol{\epsilon}_{\mathbf{q}\lambda} \cdot \mathbf{v}_b\left(\mathbf{k}+\frac{\mathbf{q}}{2}\right) \end{pmatrix}}_{M^\lambda(\mathbf{k},\mathbf{q})} \psi_{\mathbf{k}} \left(a_{\mathbf{q}\lambda} + a^\dagger_{-\mathbf{q}\lambda} \right). \tag{125}$$

Za setup od Curtisa: material 2D v xy ravnini, stoječe EM valovanje v z, vzamemo osnovni mode ($n = 1$), ki prispeva največ, v tem primeru sta dve možni polarizaciji:

$$\boldsymbol{\epsilon}_{\mathrm{TE}}(z) = (\hat{\mathbf{z}} \times \hat{\mathbf{q}}) \sin\left(\frac{\pi}{L} z\right) \tag{126}$$

$$\boldsymbol{\epsilon}_{\mathrm{TM}}(z) = \frac{\omega_0}{\omega_{\mathbf{q}}} \hat{\mathbf{q}} \sin\left(\frac{\pi}{L} z\right) + i \frac{c\, q}{\omega_{\mathbf{q}}} \hat{\mathbf{z}} \cos\left(\frac{\pi}{L} z\right) \tag{127}$$

kjer je $\mathbf{q} = q(\cos\varphi, \sin\varphi)$ valovni vektor v ravnini, $\omega_0 = \pi c/L$ in $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$. Na sredini resonatorja, pri $z = L/2$, je $\sin(\pi z/L) = 1$ in $\cos(\pi z/L) = 0$, zato:

$$\boldsymbol{\epsilon}_{\mathrm{TE}}\left(\tfrac{L}{2}\right) = \hat{\mathbf{z}} \times \hat{\mathbf{q}} = \begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix} \tag{128}$$

$$\boldsymbol{\epsilon}_{\mathrm{TM}}\left(\tfrac{L}{2}\right) = \frac{\omega_0}{\omega_{\mathbf{q}}} \hat{\mathbf{q}} = \frac{\omega_0}{\omega_{\mathbf{q}}} \begin{pmatrix} \cos\varphi \\ \sin\varphi \\ 0 \end{pmatrix} \tag{129}$$

Iz tega imamo pri $z = L/2$ možne sklopitve glede na model ($\mathbf{d} = d(\cos\theta, \sin\theta)$ v ravnini oz. $\mathbf{d} = d\,\hat{\mathbf{z}}$ izven ravnine):

Nacin	$\boldsymbol{\epsilon} \cdot \mathbf{v}_\gamma\left(\mathbf{k} + \frac{\mathbf{q}}{2}\right)$	$\boldsymbol{\epsilon} \cdot \mathbf{d}$, $\mathbf{d}$ in plane	$\boldsymbol{\epsilon} \cdot \mathbf{d}$, $\mathbf{d} \parallel \hat{\mathbf{z}}$
TE	$2t_\gamma\left(-\sin\varphi \sin\left(k_x + \frac{q_x}{2}\right) + \cos\varphi \sin\left(k_y + \frac{q_y}{2}\right)\right)$	$d\sin(\theta - \varphi)$	0
TM	$\frac{\omega_0}{\omega_{\mathbf{q}}} 2t_\gamma\left(\cos\varphi \sin\left(k_x + \frac{q_x}{2}\right) + \sin\varphi \sin\left(k_y + \frac{q_y}{2}\right)\right)$	$\frac{\omega_0}{\omega_{\mathbf{q}}} d\cos(\theta - \varphi)$	$i\,d\,\frac{c\,q}{\omega_{\mathbf{q}}} \cos\left(\frac{\pi}{L} z\right)$ $(= 0$ pri $z = \frac{L}{2})$

dipolni moment je neničeln za liho-sode kombinacije orbital in plane, tega gledamo samo glede na o.c.:

$$\mathbf{d} = \langle \phi_a | \mathbf{x} | \phi_b \rangle = \int \phi_a^*(\mathbf{x})\, \mathbf{x}\, \phi_b(\mathbf{x})\ \mathrm{d}\mathbf{x} \tag{130}$$

2.5.1 Model prostih elektronov in sklopitev s fotoni

Kaj pa če gledamo namesto tight binding, proste elektrone, torej

$$\varepsilon_{\gamma\mathbf{k}} = \pm\frac{G}{2} \mp 2t(\cos k_x + \cos k_y) \quad \text{za} \quad k_x, k_y \sim 0 \tag{131}$$

od tod

$$\varepsilon_{\gamma\mathbf{k}} \approx \pm\frac{G}{2} \mp 2t\left(1 - \frac{k_x^2}{2} + 1 - \frac{k_y^2}{2}\right) = \pm\left(\frac{G}{2} - 4t\right) \pm t(k_x^2 + k_y^2) = \pm(\varepsilon^0 + t\mathbf{k} \cdot \mathbf{k}) \tag{132}$$

Torej je Hamiltonka

$$H_e^{(0)} = \sum_{\gamma,\mathbf{k}} \left(\varepsilon_\gamma^0 + t_\gamma \mathbf{k}^2\right) c_{\gamma,\mathbf{k}}^\dagger c_{\gamma,\mathbf{k}} \tag{133}$$

Na drug način lahko poračunamo tok, ker razvijemo

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{k},\gamma=\{a,b\}} e^{i\mathbf{k}\cdot\mathbf{r}} c_{\gamma,\mathbf{k}}\,. \tag{134}$$

in za tok uporabimo

$$\mathbf{j}_e(\mathbf{r}) = -\frac{e}{2mi}\left(\psi^\dagger(\mathbf{r})\nabla\psi(\mathbf{r}) - \nabla\psi^\dagger(\mathbf{r})\psi(\mathbf{r})\right) = -\frac{e}{2mi}\frac{1}{\mathcal{V}} \sum_{\gamma,\mathbf{k},\mathbf{k}'} \left(c_{\gamma\mathbf{k}'}^\dagger c_{\gamma\mathbf{k}} e^{-i\mathbf{k}'\cdot\mathbf{r}} (i\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} - c_{\gamma\mathbf{k}'}^\dagger c_{\gamma\mathbf{k}} (-i\mathbf{k}') e^{-i\mathbf{k}'\cdot\mathbf{r}} e^{i\mathbf{k}\cdot\mathbf{r}}\right) \tag{135}$$

s FT dobimo

$$\mathbf{j}_e(-\mathbf{q}) = \int \mathrm{d}\mathbf{r}\ \mathbf{j}_e(\mathbf{r}) e^{i\mathbf{q}\cdot\mathbf{r}} = -\frac{e}{2m} \sum_{\gamma,\mathbf{k}} \left(c_{\gamma,\mathbf{k}+\mathbf{q}}^\dagger c_{\gamma\mathbf{k}}\, \mathbf{k} + (\mathbf{k} + \mathbf{q}) c_{\gamma,\mathbf{k}+\mathbf{q}}^\dagger c_{\gamma\mathbf{k}}\right) = -\frac{e}{m} \sum_{\mathbf{k}} \left(\mathbf{k} + \frac{\mathbf{q}}{2}\right) \psi_{\mathbf{k}+\mathbf{q}}^\dagger \psi_{\mathbf{k}} \tag{136}$$

kar bi dobili tudi iz bolj splošne relacije. V tem primeru dobimo enako kot Curtis, $m = 1/2t$

$$H_{e\gamma}^{(1)} = \frac{e}{m} \sum_{\mathbf{k},\mathbf{q},\lambda} \frac{1}{\sqrt{2\mathcal{V}\epsilon_0\omega_{\mathbf{q},\lambda}}} \left(\mathbf{k} + \frac{\mathbf{q}}{2}\right) \cdot \boldsymbol{\epsilon}_{\mathbf{q}\lambda}\, \psi_{\mathbf{k}+\mathbf{q}}^\dagger \psi_{\mathbf{k}} \left(a_{\mathbf{q}\lambda} + a_{-\mathbf{q}\lambda}^\dagger\right) \tag{137}$$

2.5.2 Drugi red sklopitve za t.b.

Zacnemo spet:

$$H_e^{(0)} = \sum_{\gamma,j} \varepsilon_\gamma^{(0)} c_{\gamma j}^\dagger c_{\gamma j} - \sum_{j,\boldsymbol{\delta},\gamma} t_\gamma \left(c_{\gamma j}^\dagger c_{\gamma j+\boldsymbol{\delta}} + \text{h.c.} \right) \tag{138}$$

v prisotnosti polja dobi prekrivalni integral fazo:

$$t_\gamma \to t_\gamma e^{i\theta_{j\boldsymbol{\delta}}} \qquad \theta_{j\boldsymbol{\delta}} = e \int_{\mathbf{R}_j}^{\mathbf{R}_j+\boldsymbol{\delta}} \mathbf{A}(\mathbf{r},t) \cdot d\mathbf{r} \tag{139}$$

od tod

$$H_e^{(0)}[\mathbf{A}] = \sum_{\gamma,j} \varepsilon_\gamma^{(0)} c_{\gamma j}^\dagger c_{\gamma j} - \sum_{j,\boldsymbol{\delta},\gamma} t_\gamma \left(e^{i\theta_{j\boldsymbol{\delta}}} c_{\gamma j}^\dagger c_{\gamma j+\boldsymbol{\delta}} + e^{-i\theta_{j\boldsymbol{\delta}}} c_{\gamma j+\boldsymbol{\delta}}^\dagger c_{\gamma j} \right) \tag{140}$$

od kjer razvijemo do drugega reda, prvi red bi moral biti isti kot gor (razen dipolov)

$$e^{i\theta} = 1 + \underbrace{i\theta}_{\mathcal{O}(e\mathbf{A})} \underbrace{-\frac{1}{2}\theta^2}_{\mathcal{O}(e^2A^2)} \tag{141}$$

vstavimo

$$c_{\gamma j} = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}_j} c_{\gamma\mathbf{k}} \tag{142}$$

in približek da se polje ne spremeni velko do najbližjih sosedov

$$\int_{\mathbf{R}_j}^{\mathbf{R}_j+\boldsymbol{\delta}} \mathbf{A}(\mathbf{r},t) \cdot d\mathbf{r} \approx \mathbf{A}\left(\mathbf{R}_j + \frac{\boldsymbol{\delta}}{2} \right) \cdot \boldsymbol{\delta} \qquad (\mathbf{A} \text{ se ne spremeni veliko znotraj celice+midpoint approx.}) \tag{143}$$

računamo

$$H_{e\gamma}^{(2)} = -\sum_{j,\boldsymbol{\delta},\gamma} t_\gamma \left(-\frac{\theta_{j\boldsymbol{\delta}}^2}{2} \right) \left(c_{\gamma j}^\dagger c_{\gamma j+\boldsymbol{\delta}} + c_{\gamma j+\boldsymbol{\delta}}^\dagger c_{\gamma j} \right) = \tag{144}$$

$$= \sum_{j,\boldsymbol{\delta},\gamma} t_\gamma \cdot \frac{e^2}{2} \int_{\mathbf{R}_j}^{\mathbf{R}_j+\boldsymbol{\delta}} \mathbf{A}(\mathbf{r},t) \cdot d\mathbf{r} \int_{\mathbf{R}_j}^{\mathbf{R}_j+\boldsymbol{\delta}} \mathbf{A}(\mathbf{r},t) \cdot d\mathbf{r} \left(c_{\gamma j}^\dagger c_{\gamma j+\boldsymbol{\delta}} + \text{h.c.} \right) = \tag{145}$$

$$= \sum_{j,\boldsymbol{\delta},\gamma} t_\gamma \frac{e^2}{2} \left(\sum_{\mathbf{q},\lambda} \frac{1}{\sqrt{2\varepsilon_0 \mathcal{V} \omega_{\mathbf{q}\lambda}}} \left(\boldsymbol{\epsilon}_{\mathbf{q}\lambda} \cdot \boldsymbol{\delta} \right) e^{i\mathbf{q}\cdot\left(\mathbf{R}_j + \frac{\boldsymbol{\delta}}{2}\right)} \left(a_{\mathbf{q}\lambda} + a_{-\mathbf{q}\lambda}^\dagger \right) \right) \left(\sum_{\mathbf{q}',\lambda'} \frac{1}{\sqrt{2\varepsilon_0 \mathcal{V} \omega_{\mathbf{q}'\lambda'}}} \left(\boldsymbol{\epsilon}_{\mathbf{q}'\lambda'} \cdot \boldsymbol{\delta} \right) e^{i\mathbf{q}'\cdot\left(\mathbf{R}_j + \frac{\boldsymbol{\delta}}{2}\right)} \left(a_{\mathbf{q}'\lambda'} + a_{-\mathbf{q}'\lambda'}^\dagger \right. \right. \tag{146}$$

$$\cdot \sum_{\mathbf{k},\mathbf{k}'} \frac{1}{\mathcal{V}} c_{\gamma\mathbf{k}'}^\dagger c_{\gamma\mathbf{k}} e^{-i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{R}_j} \left(e^{i\mathbf{k}\cdot\boldsymbol{\delta}} + e^{i\mathbf{k}'\cdot\boldsymbol{\delta}} \right) \tag{147}$$

ven pade ohranitev momenta za dvofotonske procese

$$\sum_j \frac{1}{\mathcal{V}} e^{i(\mathbf{q}+\mathbf{q}'-\mathbf{k}'+\mathbf{k})\cdot\mathbf{R}_j} = \delta_{\mathbf{k}',\,\mathbf{k}+\mathbf{q}+\mathbf{q}'} \tag{148}$$

kar da na koncu

$$= \sum_{\substack{\gamma,\mathbf{k},\mathbf{q},\mathbf{q}' \\ \lambda,\lambda',\boldsymbol{\delta}}} t_\gamma \frac{e^2}{2} \frac{1}{2\varepsilon_0 \mathcal{V} \sqrt{\omega_{\mathbf{q}\lambda}\omega_{\mathbf{q}'\lambda'}}} \left(\boldsymbol{\epsilon}_{\mathbf{q}\lambda} \cdot \boldsymbol{\delta} \right) \left(\boldsymbol{\epsilon}_{\mathbf{q}'\lambda'} \cdot \boldsymbol{\delta} \right) \left(a_{\mathbf{q}\lambda} + a_{-\mathbf{q}\lambda}^\dagger \right) \left(a_{\mathbf{q}'\lambda'} + a_{-\mathbf{q}'\lambda'}^\dagger \right) c_{\gamma,\mathbf{k}+\mathbf{q}+\mathbf{q}'}^\dagger c_{\gamma\mathbf{k}} \tag{149}$$

$$\cdot \left(e^{i\mathbf{k}\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}+\mathbf{q}+\mathbf{q}')\cdot\boldsymbol{\delta}} \right) \cdot e^{i(\mathbf{q}+\mathbf{q}')\cdot\frac{\boldsymbol{\delta}}{2}} \tag{150}$$

Poenostavimo zadnjifaktor v kosinus:

$$H^{(2)}_{e\gamma} = \sum_{\substack{\mathbf{k},\mathbf{q},\mathbf{q}'\\ \lambda,\lambda'}} \psi^\dagger_{\mathbf{k}+\mathbf{q}+\mathbf{q}'} M^{\lambda\lambda'}(\mathbf{k},\mathbf{q},\mathbf{q}')\psi_{\mathbf{k}} \cdot (a_{\mathbf{q},\lambda} + a^\dagger_{-\mathbf{q},\lambda})(a_{\mathbf{q}',\lambda'} + a^\dagger_{-\mathbf{q}',\lambda'}) \tag{151}$$

$$M^{\lambda\lambda'}(\mathbf{k},\mathbf{q},\mathbf{q}') = \sum_{\boldsymbol{\delta}} \frac{e^2}{2\varepsilon_0 \mathcal{V}\sqrt{\omega_{\mathbf{q},\lambda}\omega_{\mathbf{q}',\lambda'}}} (\boldsymbol{\epsilon}_{\mathbf{q},\lambda}\cdot\boldsymbol{\delta})(\boldsymbol{\epsilon}_{\mathbf{q}',\lambda'}\cdot\boldsymbol{\delta}) \cos\left(\left(\mathbf{k}+\frac{\mathbf{q}+\mathbf{q}'}{2}\right)\cdot\boldsymbol{\delta}\right)\begin{pmatrix} t_a & 0\\ 0 & t_b\end{pmatrix} \tag{152}$$

Za n.n. v square latticu eksplicitno:

$$M^{\lambda\lambda'}(\mathbf{k},\mathbf{q},\mathbf{q}') = \frac{e^2}{2\varepsilon_0 \mathcal{V}\sqrt{\omega_{\mathbf{q},\lambda}\omega_{\mathbf{q}',\lambda'}}} \left[\epsilon^x_{\mathbf{q},\lambda}\epsilon^x_{\mathbf{q}',\lambda'} \cos\left(k_x + \frac{q_x+q'_x}{2}\right) + \epsilon^y_{\mathbf{q},\lambda}\epsilon^y_{\mathbf{q}',\lambda'} \cos\left(k_y + \frac{q_y+q'_y}{2}\right)\right]\begin{pmatrix} t_a & 0\\ 0 & t_b\end{pmatrix}. \tag{153}$$

A je to consistent z $\mathcal{O}(e\mathbf{A})$? Če ja, kako dobimo dipole notri v te elemente, al se sam ne sklaplajo nikoli?

2.6 Populacijska dinamika

V splošnem sta za obe bozonski sklopitvi Hamiltonjana oblike (v prvem redu):

$$H = \sum_{\mathbf{k},\mathbf{q},\lambda} \psi^\dagger_{\mathbf{k}+\mathbf{q}} M^{\lambda}(\mathbf{k},\mathbf{q})\psi_{\mathbf{k}} \cdot (a_{\mathbf{q},\lambda} + a^\dagger_{-\mathbf{q},\lambda}) \tag{154}$$

kar prepišemo v diagonalno bazo za EI s transformacijo (32), za lžje račune definiram v diagonalni bazi:

$$d_{\gamma,\mathbf{k}} = \{\alpha_{\mathbf{k}}; \gamma = \alpha, \beta_{\mathbf{k}}; \gamma = \beta\}. \tag{155}$$

Torej je (indeks veje, oziroma polarizacije λ in vsoto bom spuščal v večini primerov):

$$H = \sum_{\substack{\mathbf{k},\mathbf{q},\lambda\\ \gamma,\gamma'}} d^\dagger_{\gamma,\mathbf{k}+\mathbf{q}} \widetilde{M}_{\gamma,\gamma'}(\mathbf{k},\mathbf{q}) d_{\gamma',\mathbf{k}} \cdot (a_{\mathbf{q},\lambda} + a^\dagger_{-\mathbf{q},\lambda}) \tag{156}$$

$$\widetilde{M}(\mathbf{k},\mathbf{q}) = U^\dagger_{\mathbf{k}+\mathbf{q}} M(\mathbf{k},\mathbf{q}) U_{\mathbf{k}} \tag{157}$$

Izpeljava je z von Neumann enačbo, da dobimo pravilne razširjene resonance, na koncu lahko primejamo s Fermijevim zlatim pravilom. Najprej prepišemo Hamiltonjan v interakcijsko sliko v diagonalni bazi je razvoj

$$H_I(t) = \sum_{\substack{\mathbf{k},\mathbf{q},\lambda\\ \gamma,\gamma'}} d^\dagger_{\gamma,\mathbf{k}+\mathbf{q}} \widetilde{M}_{\gamma,\gamma'}(\mathbf{k},\mathbf{q}) d_{\gamma',\mathbf{k}} \cdot \left(a_{\mathbf{q},\lambda}(t) + a^\dagger_{-\mathbf{q},\lambda}(t)\right) e^{i\Omega t} \tag{158}$$

$$\Omega = E_{\gamma,\mathbf{k}+\mathbf{q}} - E_{\gamma',\mathbf{k}} \tag{159}$$

Sledimo enačbi (53), kjer zamenjamo $t - s = t'$ in vzamemo $\mathrm{Tr}_S(\hat{n}_{\lambda,\mathbf{p}}\bullet)$ na obeh straneh in upoštevamo cikličnost sledi:

$$\partial_t n_{\lambda,\mathbf{p}} = -\int_0^t \mathrm{d}t' \left\langle \left[\left[d^\dagger_{\lambda,\mathbf{p}} d_{\lambda,\mathbf{p}}, H_I(t)\right], H_I(t')\right]\right\rangle \tag{160}$$

(note, $\langle\bullet\rangle = \langle\bullet\rangle_S\langle\bullet\rangle_B$) za notranji komutator rabimo:

$$[d^\dagger_A d_B, d^\dagger_C d_D] = d^\dagger_A \underbrace{\{d_B, d^\dagger_C\}}_{\delta_{BC}} d_D - d^\dagger_C \underbrace{\{d^\dagger_A, d_D\}}_{\delta_{AD}} d_B - \underbrace{\{d^\dagger_A, d^\dagger_C\}}_{0} d_B d_D + d^\dagger_C d^\dagger_A \underbrace{\{d_B, d_D\}}_{0} = d^\dagger_A d_D \delta_{BC} - d^\dagger_C d_B \delta_{AD} \tag{161}$$

oziroma:

$$[d^\dagger_A d_A, d^\dagger_C d_D] = d^\dagger_C d_D(\delta_{AC} - \delta_{AD}) \tag{162}$$

Prvi komutator je tako:

$$[n_{\lambda\mathbf{p}}, H_I(t)] = \sum_{\substack{\mathbf{k},\mathbf{q}\\ \gamma,\gamma'}} \widetilde{M}_{\gamma\gamma'}(\mathbf{k},\mathbf{q}) \overbrace{d^\dagger_{\gamma\mathbf{k}+\mathbf{q}} d_{\gamma'\mathbf{k}}}^{S} e^{i\Omega t} \overbrace{\left(a_{\mathbf{q}\lambda} + a^\dagger_{-\mathbf{q}\lambda}\right)}^{B} \left(\delta_{\lambda\gamma}\delta_{\mathbf{p},\mathbf{k}+\mathbf{q}} - \delta_{\lambda\gamma'}\delta_{\mathbf{p},\mathbf{k}}\right) \tag{163}$$

za zunanji komutator nas zanimajo samo operatorji:

$$\left[\left[n_{\lambda\mathbf{p}}, H_I(t)\right], H_I(t')\right] \sim [SB, S'B'] = SBS'B' - S'B'SB = SS'BB' - S'SB'B \tag{164}$$

oziroma bolj natančno (vzamemo še sled, ki se faktorizira za sistem in kopel)

$$\begin{aligned}
&\sum_{\substack{\mathbf{k},\mathbf{q},\mathbf{k}',\mathbf{q}'\\ \gamma,\gamma',\omega,\omega'}} \widetilde{M}_{\gamma\gamma'}(\mathbf{k},\mathbf{q})\widetilde{M}_{\omega\omega'}(\mathbf{k}',\mathbf{q}') \left(\delta_{\lambda\gamma}\delta_{\mathbf{p},\mathbf{k}+\mathbf{q}} - \delta_{\lambda\gamma'}\delta_{\mathbf{p},\mathbf{k}}\right) \cdot \\
&\quad \cdot \Bigg[\left\langle d^\dagger_{\gamma\mathbf{k}+\mathbf{q}} d_{\gamma'\mathbf{k}} d^\dagger_{\omega\mathbf{k}'+\mathbf{q}'} d_{\omega'\mathbf{k}'} \right\rangle e^{i\Omega t} e^{i\Omega' t'} \left\langle \left(a_{\mathbf{q}\lambda} + a^\dagger_{-\mathbf{q}\lambda}\right)(t) \left(a_{\mathbf{q}'\lambda'} + a^\dagger_{-\mathbf{q}'\lambda'}\right)(t') \right\rangle - \\
&\quad - \left\langle d^\dagger_{\omega\mathbf{k}'+\mathbf{q}'} d_{\omega'\mathbf{k}'} d^\dagger_{\gamma\mathbf{k}+\mathbf{q}} d_{\gamma'\mathbf{k}} \right\rangle e^{i\Omega' t'} e^{i\Omega t} \left\langle \left(a_{\mathbf{q}'\lambda'} + a^\dagger_{-\mathbf{q}'\lambda'}\right)(t') \left(a_{\mathbf{q}\lambda} + a^\dagger_{-\mathbf{q}\lambda}\right)(t) \right\rangle \Bigg],
\end{aligned} \tag{165}$$

kjer je $\Omega' = E_{\omega,\mathbf{k}'+\mathbf{q}'} - E_{\omega',\mathbf{k}'}$. Najprej bozonski deli, neničelni so samo mešani produkti (privzel sem še $\omega_{\mathbf{q}} = \omega_{-\mathbf{q}}$):

$$\begin{aligned}
\left\langle \left(a_{\mathbf{q},\lambda} + a^\dagger_{-\mathbf{q},\lambda}\right)(t) \left(a_{\mathbf{q}',\lambda'} + a^\dagger_{-\mathbf{q}',\lambda'}\right)(t') \right\rangle &= \left\langle a_{\mathbf{q},\lambda}(t) a^\dagger_{-\mathbf{q}',\lambda'}(t') \right\rangle + \left\langle a^\dagger_{-\mathbf{q},\lambda}(t) a_{\mathbf{q}',\lambda'}(t') \right\rangle = \\
&= \left\langle a_{\mathbf{q},\lambda}(t) a^\dagger_{\mathbf{q},\lambda}(t') \right\rangle + \left\langle a^\dagger_{-\mathbf{q},\lambda}(t) a_{-\mathbf{q},\lambda}(t') \right\rangle = \\
&= (N(\omega_{\mathbf{q}}) + 1) e^{-i\omega_{\mathbf{q}}(t-t')} + N(\omega_{\mathbf{q}}) e^{i\omega_{\mathbf{q}}(t-t')}
\end{aligned} \tag{166}$$

kjer mora veljati $\mathbf{q}' = -\mathbf{q}$ in $\lambda = \lambda'$ (note, ta lambda ni isti kot un fermionski lambda, zmanjkuje mi lepih grskih crk za pisat). Podobno je z drugim členom. Uvedemo še $\tau = t - t'$, integral se ne spremeni z uvedbo te spremenljivke. A la Wick naredimo še za fermionske dele (ker so d diagonalni lahko uporabimo Wicka):

$$\langle d^\dagger_A d_B d^\dagger_C d_D \rangle = \langle d^\dagger_A d_B \rangle \langle d^\dagger_C d_D \rangle + \langle d^\dagger_A d_D \rangle \langle d_B d^\dagger_C \rangle = \tag{167}$$

$$= n_A n_C \delta_{AB}\delta_{CD} + n_A(1 - n_C)\delta_{BC}\delta_{AD} \tag{168}$$

$$\hookrightarrow \delta_{AB} \mapsto \delta_{\gamma\gamma'}\delta_{\mathbf{k}+\mathbf{q},\mathbf{k}} = 0 \quad \text{odpade zaradi razlike delta funkcij} \tag{169}$$

oziroma (prvi člen je nič, ker se nič ne zgodi/ ni sipanje/ga ignoriramo, ker smo fiziki), upoštevamo še $\mathbf{q}' = -\mathbf{q}$:

$$\left\langle d^\dagger_{\gamma\mathbf{k}+\mathbf{q}} d_{\gamma'\mathbf{k}} d^\dagger_{\omega\mathbf{k}'+\mathbf{q}'} d_{\omega'\mathbf{k}'} \right\rangle = n_{\gamma,\mathbf{k}+\mathbf{q}} \left(1 - n_{\omega,\mathbf{k}'-\mathbf{q}}\right) \delta_{\gamma'\omega}\delta_{\mathbf{k},\mathbf{k}'-\mathbf{q}}\delta_{\gamma\omega'} \tag{170}$$

$$\left\langle d^\dagger_{\omega\mathbf{k}'+\mathbf{q}'} d_{\omega'\mathbf{k}'} d^\dagger_{\gamma\mathbf{k}+\mathbf{q}} d_{\gamma'\mathbf{k}} \right\rangle = n_{\omega,\mathbf{k}'-\mathbf{q}} \left(1 - n_{\gamma,\mathbf{k}+\mathbf{q}}\right) \delta_{\omega'\gamma}\delta_{\mathbf{k}',\mathbf{k}+\mathbf{q}}\delta_{\omega\gamma'} \tag{171}$$

iz delt sledi za eksponenta/časovni razvoj v obeh clenih:

$$\Omega' = E_{\omega,\mathbf{k}'+\mathbf{q}'} - E_{\omega',\mathbf{k}'} = E_{\gamma',\mathbf{k}} - E_{\gamma,\mathbf{k}+\mathbf{q}} = -\Omega \implies e^{i\Omega' t'} e^{i\Omega t} = e^{i\Omega\tau} \tag{172}$$

Vse skupaj je treba še integrirati, dodamo regulator, da pole izmaknemo iz osi:

$$\int_0^\infty e^{i\tau(\Omega \pm \omega_{\mathbf{q}} + i\gamma)}\, d\tau = -\frac{i e^{i(\Omega\pm\omega_{\mathbf{q}})\tau} e^{-\gamma\tau}}{\Omega \pm \omega_{\mathbf{q}} + i\gamma}\Bigg|_0^\infty = \frac{i}{\Omega \pm \omega_{\mathbf{q}} + i\gamma} = D_\pm(\Omega) \tag{173}$$

oziroma skupaj:

$$\partial_t n_{\lambda,\mathbf{p}} = -\int_0^{t\to\infty} \sum_{\substack{\mathbf{k},\mathbf{q}\\ \gamma,\gamma'}} \left(\delta_{\lambda\gamma}\delta_{\mathbf{p},\mathbf{k}+\mathbf{q}} - \delta_{\lambda\gamma'}\delta_{\mathbf{p},\mathbf{k}}\right) \cdot \tag{174}$$

$$\cdot \Big(n_{\gamma,\mathbf{k}+\mathbf{q}}(1 - n_{\gamma',\mathbf{k}}) \widetilde{M}_{\gamma\gamma'}(\mathbf{k},\mathbf{q}) \widetilde{M}_{\gamma'\gamma}(\mathbf{k}+\mathbf{q},-\mathbf{q}) e^{i\Omega\tau} \left((N(\omega_{\mathbf{q}}) + 1) e^{-i\omega_{\mathbf{q}}\tau} + N(\omega_{\mathbf{q}}) e^{i\omega_{\mathbf{q}}\tau}\right) \tag{175}$$

$$- n_{\gamma',\mathbf{k}}(1 - n_{\gamma,\mathbf{k}+\mathbf{q}}) \widetilde{M}_{\gamma\gamma'}(\mathbf{k},\mathbf{q}) \widetilde{M}_{\gamma'\gamma}(\mathbf{k}+\mathbf{q},-\mathbf{q}) e^{i\Omega\tau} \left((N(\omega_{\mathbf{q}}) + 1) e^{i\omega_{\mathbf{q}}\tau} + N(\omega_{\mathbf{q}}) e^{-i\omega_{\mathbf{q}}\tau}\right) \Big)\, d\tau \tag{176}$$

Iz hermitskosti sledi:

$$\widetilde{M}_{\gamma'\gamma}(\mathbf{k}+\mathbf{q},-\mathbf{q}) = \widetilde{M}_{\gamma'\gamma}(\mathbf{k},\mathbf{q})^* \tag{177}$$

ko razpišemo vseh 8 členov z deltami in preimenujemo indekse ostane ($\Omega = E_{\lambda,\mathbf{p}} - E_{\gamma,\mathbf{k}}$)

$$\partial_t n_{\lambda,\mathbf{p}} = -\sum_{\mathbf{k},\gamma} \left|\widetilde{M}_{\gamma\lambda}\right|^2 n_{\lambda\mathbf{p}}(1-n_{\gamma\mathbf{k}}) \left\{(N_{\mathbf{p}-\mathbf{k}}+1)\left[D_-(\Omega)+D_+(-\Omega)\right] + N_{\mathbf{p}-\mathbf{k}}\left[D_+(\Omega)+D_-(-\Omega)\right]\right\} + \tag{178}$$

$$+\sum_{\mathbf{k},\gamma} \left|\widetilde{M}_{\lambda\gamma}\right|^2 (1-n_{\lambda\mathbf{p}}) n_{\gamma\mathbf{k}} \left\{(N_{\mathbf{p}-\mathbf{k}}+1)\left[D_+(\Omega)+D_-(-\Omega)\right] + N_{\mathbf{p}-\mathbf{k}}\left[D_-(\Omega)+D_+(-\Omega)\right]\right\} \tag{179}$$

od kjer lahko poenostavimo:

$$D_-(\Omega)+D_+(-\Omega) = \frac{2\gamma}{(\Omega-\omega_{\mathbf{q}})^2+\gamma^2} \equiv \mathcal{A}_{\mathbf{q}}(\Omega) \tag{180}$$

$$D_+(\Omega)+D_-(-\Omega) = \frac{2\gamma}{(\Omega+\omega_{\mathbf{q}})^2+\gamma^2} = \mathcal{A}_{\mathbf{q}}(-\Omega) \tag{181}$$

oziroma končno:;

$$\partial_t n_{\gamma\mathbf{k}} = \sum_{\mathbf{k}',\gamma'} \Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} n_{\gamma'\mathbf{k}'}(1-n_{\gamma\mathbf{k}}) - \Gamma_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'} n_{\gamma\mathbf{k}}(1-n_{\gamma'\mathbf{k}'}) \tag{182}$$

$$\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} = \sum_{\lambda} \left|\tilde{M}^{\lambda}_{\gamma\gamma'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')\right|^2 \cdot \left\{(N_\lambda(-\Omega)+1)\mathcal{A}_{\mathbf{k}-\mathbf{k}',\lambda}(-\Omega) + N_\lambda(\Omega)\mathcal{A}_{\mathbf{k}-\mathbf{k}',\lambda}(\Omega)\right\} \tag{183}$$

$$\Gamma_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'} = \sum_{\lambda} \left|\tilde{M}^{\lambda}_{\gamma\gamma'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')\right|^2 \cdot \left\{(N_\lambda(\Omega)+1)\mathcal{A}_{\mathbf{k}'-\mathbf{k},\lambda}(\Omega) + N_\lambda(-\Omega)\mathcal{A}_{\mathbf{k}'-\mathbf{k},\lambda}(-\Omega)\right\} \tag{184}$$

$$\Omega = E_{\gamma\mathbf{k}} - E_{\gamma'\mathbf{k}'}, \qquad N(-\Omega) = -1 - N(\Omega) \tag{185}$$

preverimo predznake v tabeli 1.

Dobimo enako kot Curtis, čeprav smo fotone obravnavali v približku: za korelatorje oziroma produkte bozonov smo privzeli da imajo točno definirane resonance in dodali regulator na roke. V resnici so fotoni sklopljeni z okolico, kar jim da končno širino v disperziji. Rate matrike lahko še porenostavimo na razlike Lotentzovih vrhov:

$$\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} = |\widetilde{M}_{\gamma\gamma'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2 N(\Omega) \cdot \left\{\mathcal{A}_{\mathbf{k}-\mathbf{k}'}(\Omega) - \mathcal{A}_{\mathbf{k}-\mathbf{k}'}(-\Omega)\right\} \tag{186}$$

$$\Gamma_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'} = |\widetilde{M}_{\gamma\gamma'}(\mathbf{k}',\mathbf{k}-\mathbf{k}')|^2 (N(\Omega)+1) \cdot \left\{\mathcal{A}_{\mathbf{k}'-\mathbf{k}}(\Omega) - \mathcal{A}_{\mathbf{k}'-\mathbf{k}}(-\Omega)\right\} \tag{187}$$

od kjer vidimo, da s tem točno zadostimo detaljnemu ravnovesju(?nevem kaj je prevod):

$$\frac{\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}}}{\Gamma_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'}} = \frac{N(\Omega)}{N(\Omega)+1} = e^{-\Omega/T}\,. \tag{188}$$

Če vzamemo GGE obliko, za boljšo stabilnost: $n_{\gamma\mathbf{k}} = 1/(1+\exp(\mu_{\gamma\mathbf{k}}))$, potem je

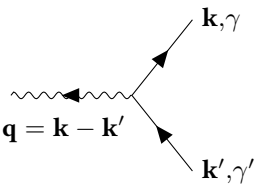
$$\partial_t n_{\gamma\mathbf{k}} = -n_{\gamma\mathbf{k}}(1-n_{\gamma\mathbf{k}})\partial_t \mu_{\gamma\mathbf{k}} = \mathcal{I}_{\gamma\mathbf{k}} \tag{189}$$

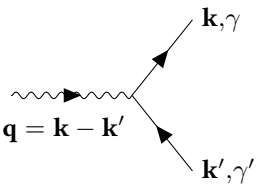
$$\mathcal{I}_{\gamma\mathbf{k}} = \sum_{\gamma',\mathbf{k}'} \left[\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}}(1-n_{\gamma\mathbf{k}})n_{\gamma'\mathbf{k}'} - \Gamma_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'}(1-n_{\gamma'\mathbf{k}'})n_{\gamma\mathbf{k}}\right] \tag{190}$$

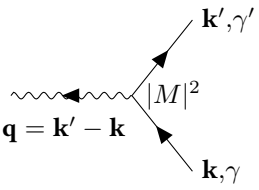
$$\partial_t \mu_{\gamma\mathbf{k}} = -\frac{\mathcal{I}_{\gamma\mathbf{k}}}{n_{\gamma\mathbf{k}}(1-n_{\gamma\mathbf{k}})} \tag{191}$$

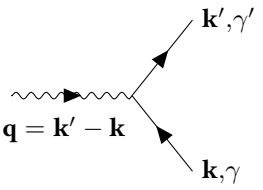
oziroma:

$$\partial_t \mu_{\gamma\mathbf{k}} = -\sum_{\gamma',\mathbf{k}',b} \Gamma^b_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} \frac{1+e^{\mu_{\gamma\mathbf{k}}}}{1+e^{\mu_{\gamma'\mathbf{k}'}}} - \Gamma^b_{\gamma\mathbf{k}\to\gamma'\mathbf{k}'} \frac{1+e^{-\mu_{\gamma\mathbf{k}}}}{1+e^{-\mu_{\gamma'\mathbf{k}'}}}\,, \tag{192}$$

<table>
<tr>
<td>A1
 

$\Omega + \omega_{\mathbf{q}} \simeq 0$
$= E_{\gamma\mathbf{k}} - E_{\gamma'\mathbf{k}'} + \omega_{\mathbf{q}}$
$= \underbrace{(E_{\gamma\mathbf{k}} + \omega_{\mathbf{q}})}_{E_f} - \underbrace{E_{\gamma'\mathbf{k}'}}_{E_i}.$</td>
<td>A2
 

$\Omega - \omega_{\mathbf{q}} \simeq 0$
$= E_{\gamma\mathbf{k}} - E_{\gamma'\mathbf{k}'} - \omega_{\mathbf{q}}$
$= \underbrace{E_{\gamma\mathbf{k}}}_{E_f} - \underbrace{(E_{\gamma'\mathbf{k}'} + \omega_{\mathbf{q}})}_{E_i}.$</td>
</tr>
<tr>
<td>B1
 

$\Omega - \omega_{\mathbf{q}} \simeq 0$
$= E_{\gamma\mathbf{k}} - E_{\gamma'\mathbf{k}'} - \omega_{\mathbf{q}}$
$= -\big[(E_{\gamma'\mathbf{k}'} + \omega_{\mathbf{q}}) - E_{\gamma\mathbf{k}}\big].$</td>
<td>B2
 

$\Omega + \omega_{\mathbf{q}} \simeq 0$
$= E_{\gamma\mathbf{k}} - E_{\gamma'\mathbf{k}'} + \omega_{\mathbf{q}}$
$= -\big[E_{\gamma'\mathbf{k}'} - (E_{\gamma\mathbf{k}} + \omega_{\mathbf{q}})\big].$</td>
</tr>
</table>

Tabela 1 | Feynman diagrami za vse štiri procese (iz in v fiskno stanje $\gamma, \mathbf{k}$).

kjer je $\Gamma^b = \Gamma(T_b)$. Izraz se poenostavi če upoštevamo še zahtevo po d.b.:

$$\partial_t \mu_{\gamma\mathbf{k}} = \sum_{\gamma'\mathbf{k}',b} \Gamma^b_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} \frac{1+e^{\mu_{\gamma\mathbf{k}}}}{1+e^{\mu_{\gamma'\mathbf{k}'}}} \left(e^{\Omega/T_b + \mu_{\gamma'\mathbf{k}'} - \mu_{\gamma\mathbf{k}}} - 1\right) \tag{193}$$

2.7 Sprememba koordinat

Za 2D square lattice mrežo uvedemo:

$$s = \cos k_x + \cos k_y, \qquad u = \cos k_x - \cos k_y. \tag{194}$$

Obratna transformacija je bijektivna na irreducible trikotniku ($\Gamma - X - M$, oziroma: $\{(k_x,k_y) : 0 \le k_y \le k_x \le \pi\}$):

$$\mathbf{k}(s,u) = \left(\arccos\left(\frac{s+u}{2}\right), \arccos\left(\frac{s-u}{2}\right)\right)^{\mathsf{T}} \tag{195}$$

Z nekaj obračanja dobimo območje kjer sta definirani novi spremenljivki:

$$-2 \le s \le 2, \qquad |s| - 2 \le u \le 0 \tag{196}$$

$$\varepsilon_a(s) = \frac{G}{2} - 2t_a s, \qquad \varepsilon_b(s) = -\frac{G}{2} - 2t_b s. \tag{197}$$

$$\varepsilon_a^H(s) = \frac{G}{2} - 2t_a s + \frac{V}{2}(n-m), \tag{198}$$

$$\varepsilon_b^H(s) = -\frac{G}{2} - 2t_b s + \frac{V}{2}(n+m). \tag{199}$$

$$\eta(s) = \frac{\varepsilon_a^H(s) + \varepsilon_b^H(s)}{2} = \frac{Vn}{2} - (t_a + t_b)s, \tag{200}$$

$$\xi(s) = \frac{\varepsilon_a^H(s) - \varepsilon_b^H(s)}{2} = \frac{G - Vm}{2} - (t_a - t_b)s. \tag{201}$$

$$E_{\alpha,\beta}(s,u) = \frac{Vn}{2} - (t_a + t_b)s \pm \sqrt{\left[\frac{G - Vm}{2} - (t_a - t_b)s\right]^2 + |\Delta|^2}. \tag{202}$$

$$u_{\mathbf{k}} \to u^2(s) = \frac{1}{2}\left(1 + \frac{\xi(s)}{\sqrt{\xi^2(s) + \Delta^2}}\right), \tag{203}$$

$$v_{\mathbf{k}} \to v^2(s) = \frac{1}{2}\left(1 - \frac{\xi(s)}{\sqrt{\xi^2(s) + \Delta^2}}\right), \tag{204}$$

$$u_{\mathbf{k}} v_{\mathbf{k}} \to u(s)v(s) = -\frac{\Delta}{2\sqrt{\xi^2(s) + \Delta^2}}. \tag{205}$$

Za integrale/fazne vsote po 1. B.C. je Jakobijan

$$\left|\frac{\partial(s,u)}{\partial(k_x,k_y)}\right| = 2\sin k_x \sin k_y. \tag{206}$$

Za intergal po 1. B.C. lahko v najbolj simetričnem primeru (za funckijo $F(g\mathbf{k}) = F(\mathbf{k})$,$\forall g \in C_{4v}$) pišemo (upoštevamo integral po trikotniku $\cdot 8$)

$$\langle F \rangle_{\text{B.C.}} = \frac{1}{(2\pi)^2}\int_{\text{B.C.}} d^2k\, F(\mathbf{k}) = \frac{1}{\pi^2}\int_{-2}^{2} ds \int_{|s|-2}^{0} du\, \frac{F(s,u)}{\sqrt{1 - \frac{(s+u)^2}{4}}\sqrt{1 - \frac{(s-u)^2}{4}}}. \tag{207}$$

Velikokrat je $F(s,u) = F(s)$, takrat velja:

$$\pi^2 g(s) = \int_{|s|-2}^{0} du \frac{1}{\sqrt{1 - \frac{(s+u)^2}{4}}\sqrt{1 - \frac{(s-u)^2}{4}}} = \frac{4}{2+|s|} K\left(\frac{2-|s|}{2+|s|}\right) \tag{208}$$

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2\sin^2\theta}} \tag{209}$$

okoli $s \approx 0$:

$$\pi^2 g(s) \approx (2 - |s|) \cdot \frac{1}{2}\ln\left(\frac{8}{|s|}\right) \approx \ln\left(\frac{8}{|s|}\right) \tag{210}$$

$$\langle F \rangle_{\text{B.C.}} = \int_{-2}^{2} g(s) \cdot F(s)\, ds \quad ; \quad g(s) = \frac{1}{\pi^2}\frac{4}{2+|s|} K\left(\frac{2-|s|}{2+|s|}\right) \tag{211}$$

V termičnem primeru velja:

$$\Delta = V\left\langle \frac{\Delta}{2\sqrt{\xi^2(s) + \Delta^2}}[f_\beta(s) - f_\alpha(s)] \right\rangle_{\text{B.C.}}, \tag{212}$$

$$m = \left\langle \frac{\xi(s)}{\sqrt{\xi^2(s) + \Delta^2}}[f_\alpha(s) - f_\beta(s)] \right\rangle_{\text{B.C.}}, \tag{213}$$

$$n = \langle f_\alpha(s) + f_\beta(s) \rangle_{\text{B.C.}}. \tag{214}$$

Za najbolj splošen primer je integral enak:

$$\langle F(\mathbf{k})\rangle_{\text{B.C.}} = = \frac{1}{8\pi^2}\sum_{g\in C_{4v}}\int_{-2}^{2} ds \int_{|s|-2}^{0} du\, \frac{F(g\mathbf{k}(s,u))}{\sqrt{1-\frac{(s+u)^2}{4}}\sqrt{1-\frac{(s-u)^2}{4}}}\,. \tag{215}$$

Najprej opazimo da je za $g \in C_{4v}$ spremeljivka s in vse količine, ki so samo funkcije s invariantna, spremeljivka u pa se transfromira kot $u \to u\,, g = \{E, C_2, \sigma_v, \sigma_{v'}\}$ in $u \to -u$ za ostale 4 elemente. Torej, če je $F(s,u)$ soda v u je invariantna, v splošnem pa je:

$$\sum_{g\in C_{4v}} F(g\mathbf{k}) = 4F(s,u) + 4F(s,-u)\,. \tag{216}$$

zato je v splošnem:

$$\langle F(\mathbf{k})\rangle_{\text{B.C.}} = \frac{1}{2\pi^2}\int_{-2}^{2} ds \int_{|s|-2}^{0} du\, \frac{F(s,u)+F(s,-u)}{\sqrt{1-\frac{(s+u)^2}{4}}\sqrt{1-\frac{(s-u)^2}{4}}}\,. \tag{217}$$

Za netermični primer je rahla komplikacija z $f_\gamma(s) \mapsto n_\gamma(s,u)$. Prepisati moramo sipalni integral $\mathcal{I}_{\gamma\mathbf{k}}$ na nove spremenljivke.

$$\partial_t \mu_{\gamma\mathbf{k}} = \sum_{\gamma',b}\frac{1}{2\pi^2}\int_{-2}^{2} ds' \int_{|s|-2}^{0} du'\, J(s',u') \sum_{i=1}^{2}\left[\Gamma^b_{\gamma'\mathbf{k}'_i\to\gamma\mathbf{k}}\frac{1+e^{\mu_{\gamma\mathbf{k}}}}{1+e^{\mu_{\gamma'\mathbf{k}'_i}}}\left(e^{\Omega/T_b+\mu_{\gamma'\mathbf{k}'_i}-\mu_{\gamma\mathbf{k}}}-1\right)\right] \tag{218}$$

$$\Gamma^b_{\gamma'\mathbf{k}'_i\to\gamma\mathbf{k}} = |\widetilde{M}_{\gamma\gamma'}(\mathbf{k}'_i, \mathbf{k}-\mathbf{k}'_i)|^2 N(\Omega)\cdot\left\{\mathcal{A}_{\mathbf{k}-\mathbf{k}'_i}(\Omega) - \mathcal{A}_{\mathbf{k}-\mathbf{k}'_i}(-\Omega)\right\} \tag{219}$$

$$\mathbf{k}'_1 = \mathbf{k}'(s',u')\,, \quad \mathbf{k}'_2 = \mathbf{k}'(s',-u') \tag{220}$$

kjer

$$\mathbf{k}'_1 = \left(k'_{x1},\, k'_{y1}\right) = \left(\arccos\left(\frac{s'+u'}{2}\right),\, \arccos\left(\frac{s'-u'}{2}\right)\right) \tag{221}$$

$$\mathbf{k}'_2 = \left(k'_{x2},\, k'_{y2}\right) = \left(\arccos\left(\frac{s'-u'}{2}\right),\, \arccos\left(\frac{s'+u'}{2}\right)\right) \tag{222}$$

če je Γ popolnoma simetrična na vse operacije vsaj vemo, da je $\mu_\gamma(s,u) = \mu_\gamma(s,-u)$, ne določa pa to odvisnosti od u. Recimo, da fiksiramo $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2q^2}$. Ta je invariantna na C_{4v}, tudi $\Omega = \Omega(s)$ je, zato je $\mathcal{A}_{\mathbf{q}}(\Omega)$ invariantna. Enako velja za Bose faktorje.

3 Konkretna implementacija za sklopitev s fotoni

konkretna implementacija/racun za verteks za fotone:

$$\mathbf{q} = q(\cos\varphi, \sin\varphi), \qquad \varphi = \text{atan2}(q_y,q_x)\ (\varphi = 0 \text{ pri } \mathbf{q} = 0), \qquad \omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2q^2}, \tag{223}$$

$$s_{\mathbf{k}} = \cos k_x + \cos k_y, \qquad \varepsilon_{a,\mathbf{k}} = \tfrac{G}{2} - 2t_a s_{\mathbf{k}}, \qquad \varepsilon_{b,\mathbf{k}} = -\tfrac{G}{2} - 2t_b s_{\mathbf{k}}. \tag{224}$$

$$\mathbf{v}_\gamma(\mathbf{p}) = \nabla_{\mathbf{p}}\,\varepsilon_{\gamma,\mathbf{p}} = \nabla_{\mathbf{p}}\left[\pm\tfrac{G}{2} - 2t_\gamma(\cos p_x + \cos p_y)\right] = 2t_\gamma\,(\sin p_x,\ \sin p_y). \tag{225}$$

$$\mathbf{v}_\gamma\big(\mathbf{k}' + \tfrac{\mathbf{q}}{2}\big) = 2t_\gamma\,(\sin\big(k'_x + \tfrac{q_x}{2}\big)\,,\ \sin\big(k'_y + \tfrac{q_y}{2}\big)), \tag{226}$$

$$\boldsymbol{\epsilon}_\lambda \cdot \mathbf{v}_\gamma\big(\mathbf{k}' + \tfrac{\mathbf{q}}{2}\big) = 2t_\gamma\, X_\lambda, \qquad X_\lambda = \boldsymbol{\epsilon}_\lambda \cdot (\sin\big(k'_x + \tfrac{q_x}{2}\big)\,,\ \sin\big(k'_y + \tfrac{q_y}{2}\big)). \tag{227}$$

$$\boldsymbol{\epsilon}_{\text{TE}} = (-\sin\varphi,\ \cos\varphi), \qquad \boldsymbol{\epsilon}_{\text{TM}} = \frac{\omega_0}{\omega_{\mathbf{q}}}(\cos\varphi,\ \sin\varphi), \qquad \mathbf{d} = d\,(1{,}0). \tag{228}$$

$$X_{\mathrm{TE}} = -\sin\varphi\, h_x + \cos\varphi\, h_y, \qquad X_{\mathrm{TM}} = \frac{\omega_0}{\omega_{\mathbf{q}}}\left(\cos\varphi\, h_x + \sin\varphi\, h_y\right), \tag{229}$$

$$P_{\mathrm{TE}} = -d\sin\varphi, \qquad P_{\mathrm{TM}} = \frac{\omega_0}{\omega_{\mathbf{q}}}\, d\cos\varphi, \tag{230}$$

$$\Delta_{ab} = \varepsilon_{a,\mathbf{k}} - \varepsilon_{b,\mathbf{k}'} = G - 2t_a s_{\mathbf{k}} + 2t_b s_{\mathbf{k}'}, \qquad \Delta_{ba} = \varepsilon_{b,\mathbf{k}} - \varepsilon_{a,\mathbf{k}'} = -G - 2t_b s_{\mathbf{k}} + 2t_a s_{\mathbf{k}'}. \tag{231}$$

$$M^{\lambda}(\mathbf{k}',\mathbf{q}) = \sqrt{\frac{A}{\omega_{\mathbf{q}}}}\begin{pmatrix} 2t_a X_\lambda & iP_\lambda \Delta_{ab} \\ iP_\lambda \Delta_{ba} & 2t_b X_\lambda \end{pmatrix} \tag{232}$$

$$\widetilde{M}^{\lambda} = U_{\mathbf{k}}^{T} M^{\lambda} U_{\mathbf{k}'}, \qquad U_{\mathbf{k}} = \begin{pmatrix} u & v \\ -v & u \end{pmatrix}, \qquad u = \sqrt{\tfrac{1}{2}\left(1 + \tfrac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right)}, \quad v = \mathrm{sgn}(\Delta)\sqrt{\tfrac{1}{2}\left(1 - \tfrac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right)}. \tag{233}$$

$$\widetilde{M}^{\lambda}_{\alpha\alpha} = \sqrt{\frac{A}{\omega_{\mathbf{q}}}}\left[X_\lambda\left(2t_a uu' + 2t_b vv'\right) - iP_\lambda\left(uv'\Delta_{ab} + vu'\Delta_{ba}\right)\right], \tag{234}$$

$$\widetilde{M}^{\lambda}_{\alpha\beta} = \sqrt{\frac{A}{\omega_{\mathbf{q}}}}\left[X_\lambda\left(2t_a uv' - 2t_b vu'\right) + iP_\lambda\left(uu'\Delta_{ab} - vv'\Delta_{ba}\right)\right], \tag{235}$$

$$\widetilde{M}^{\lambda}_{\beta\alpha} = \sqrt{\frac{A}{\omega_{\mathbf{q}}}}\left[X_\lambda\left(2t_a vu' - 2t_b uv'\right) + iP_\lambda\left(uu'\Delta_{ba} - vv'\Delta_{ab}\right)\right], \tag{236}$$

$$\widetilde{M}^{\lambda}_{\beta\beta} = \sqrt{\frac{A}{\omega_{\mathbf{q}}}}\left[X_\lambda\left(2t_a vv' + 2t_b uu'\right) + iP_\lambda\left(vu'\Delta_{ab} + uv'\Delta_{ba}\right)\right]. \tag{237}$$

$$\sum_{\lambda}\left|\widetilde{M}^{\lambda}_{\gamma\gamma'}\right|^2 = \frac{A}{\omega_{\mathbf{q}}}\left[\left(X_{\mathrm{TE}}^2 + X_{\mathrm{TM}}^2\right)C_{\gamma\gamma'}^2 + \left(\sin^2\varphi + \frac{\omega_0^2}{\omega_{\mathbf{q}}^2}\cos^2\varphi\right)D_{\gamma\gamma'}^2\right], \tag{238}$$

$$C_{\alpha\alpha} = 2t_a uu' + 2t_b vv', \qquad D_{\alpha\alpha} = d\left(uv'\Delta_{ab} + vu'\Delta_{ba}\right), \tag{239}$$

$$C_{\alpha\beta} = 2t_a uv' - 2t_b vu', \qquad D_{\alpha\beta} = d\left(uu'\Delta_{ab} - vv'\Delta_{ba}\right), \tag{240}$$

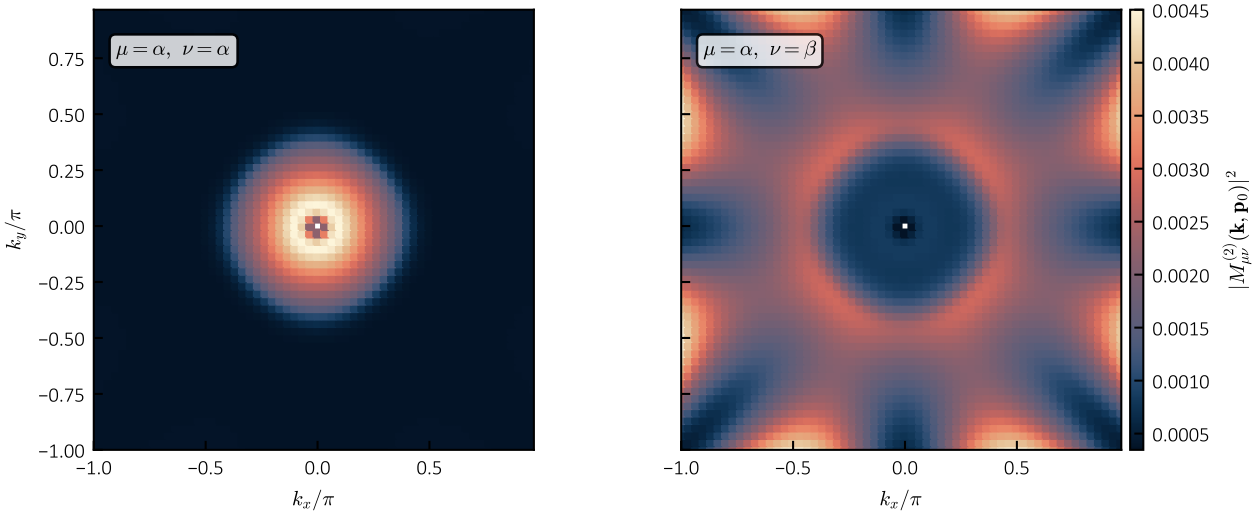
$$C_{\beta\alpha} = 2t_a vu' - 2t_b uv', \qquad D_{\beta\alpha} = d\left(uu'\Delta_{ba} - vv'\Delta_{ab}\right), \tag{241}$$

$$C_{\beta\beta} = 2t_a vv' + 2t_b uu', \qquad D_{\beta\beta} = d\left(vu'\Delta_{ab} + uv'\Delta_{ba}\right). \tag{242}$$

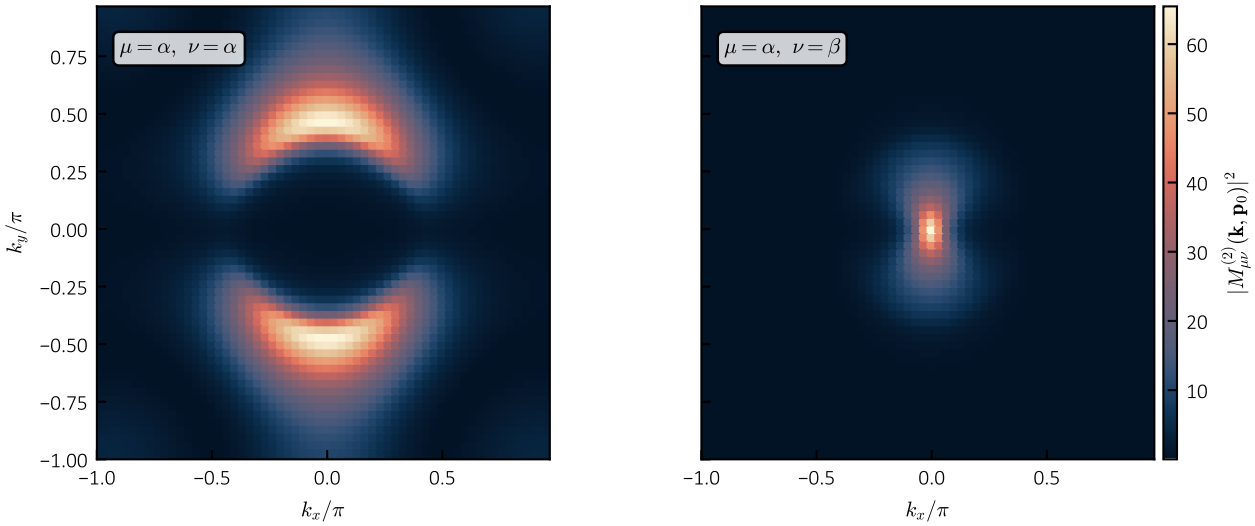
$$\begin{aligned}
\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} &= \sum_{\lambda}\left|\widetilde{M}^{\lambda}_{\gamma\gamma'}\right|^2 N(\Omega)\,\frac{8\gamma\,\omega_{\mathbf{q}}\,\Omega}{\left[(\Omega-\omega_{\mathbf{q}})^2+\gamma^2\right]\left[(\Omega+\omega_{\mathbf{q}})^2+\gamma^2\right]} = \\
&= A\Bigg[\Bigg(\left(\sin^2\varphi + \frac{\omega_0^2}{\omega_{\mathbf{q}}^2}\cos^2\varphi\right)\sin^2\left(k_x' + \tfrac{q_x}{2}\right) + \left(\cos^2\varphi + \frac{\omega_0^2}{\omega_{\mathbf{q}}^2}\sin^2\varphi\right)\sin^2\left(k_y' + \tfrac{q_y}{2}\right) + \\
&+ 2\left(\frac{\omega_0^2}{\omega_{\mathbf{q}}^2} - 1\right)\cos(\varphi)\sin(\varphi)\sin\left(k_x' + \tfrac{q_x}{2}\right)\sin\left(k_y' + \tfrac{q_y}{2}\right)\Bigg)C_{\gamma\gamma'}^2 + \\
&+ \left(\sin^2\varphi + \frac{\omega_0^2}{\omega_{\mathbf{q}}^2}\cos^2\varphi\right)D_{\gamma\gamma'}^2\Bigg]\cdot\Omega N(\Omega)\cdot\frac{8\gamma}{\left[(\Omega-\omega_{\mathbf{q}})^2+\gamma^2\right]\left[(\Omega+\omega_{\mathbf{q}})^2+\gamma^2\right]}.
\end{aligned} \tag{243}$$

4 Vizualizacije kopeli za fotone

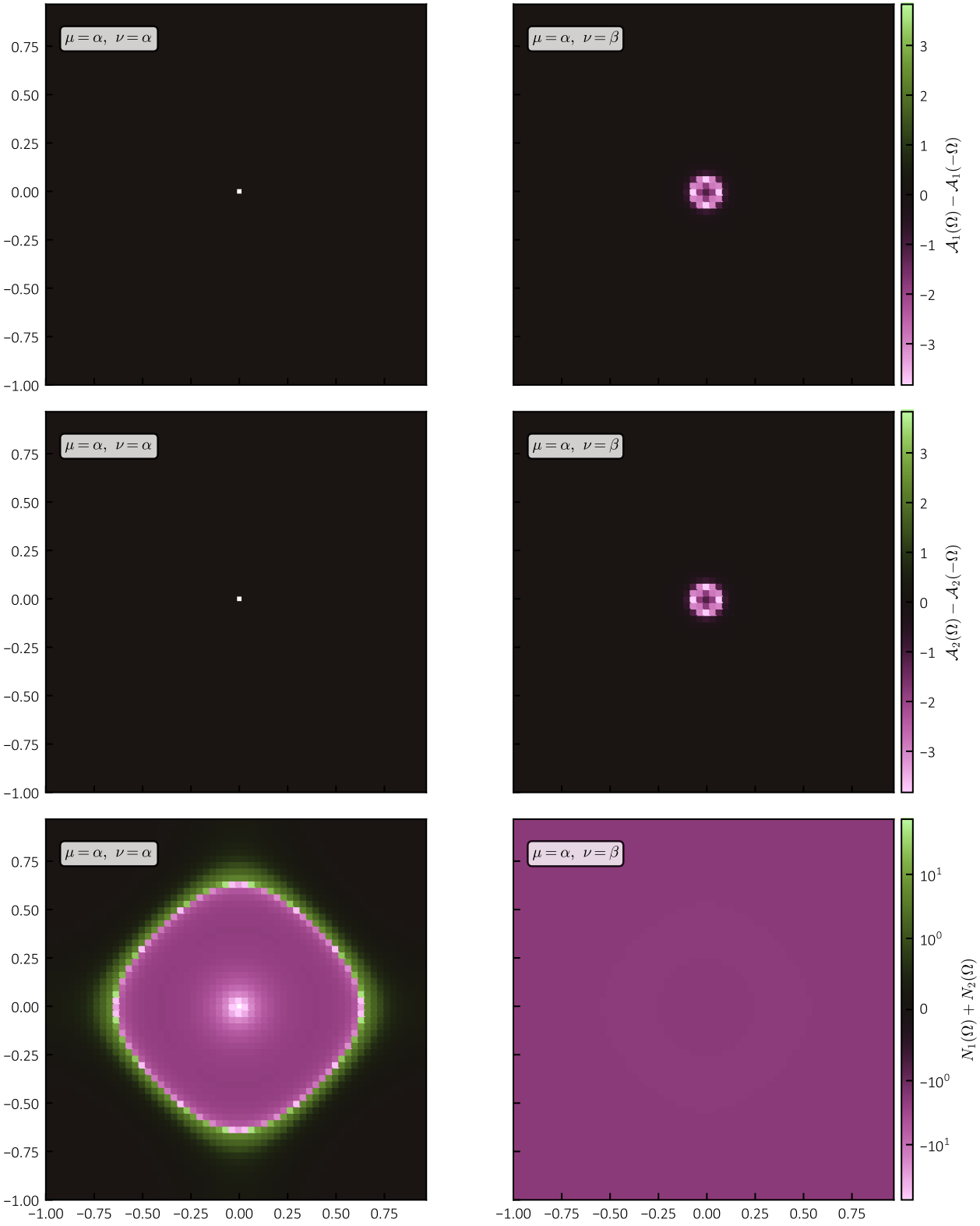
Vsi naslednji grafi so v tocki $T_1 = 1.2T_c, T_2 = 0.2T_c$ in uporabljajo fotonski verteks, kot opisano zgoraj, $\mathbf{q} = \mathbf{k} - \mathbf{p}_0$.



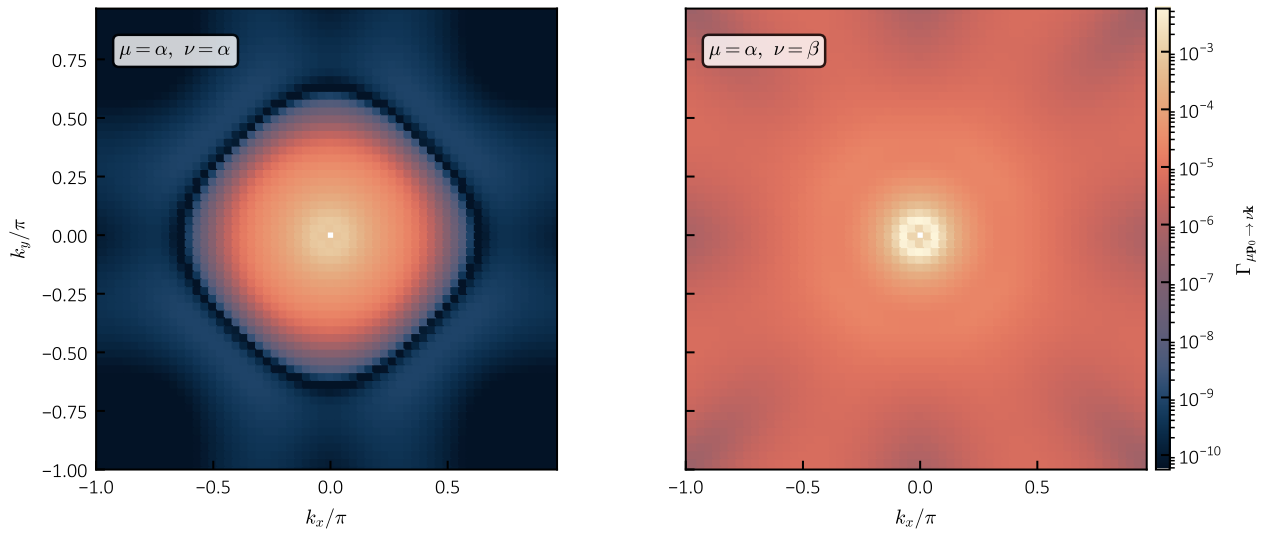
Slika 2 | Verteks pri $\mathbf{d} = 0$ za $\mathbf{p}_0 = (0,0)$ za obe kopeli z disperzijo: $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$, indeks gor je indeks kopeli (za obe je tu enaka disperzija).



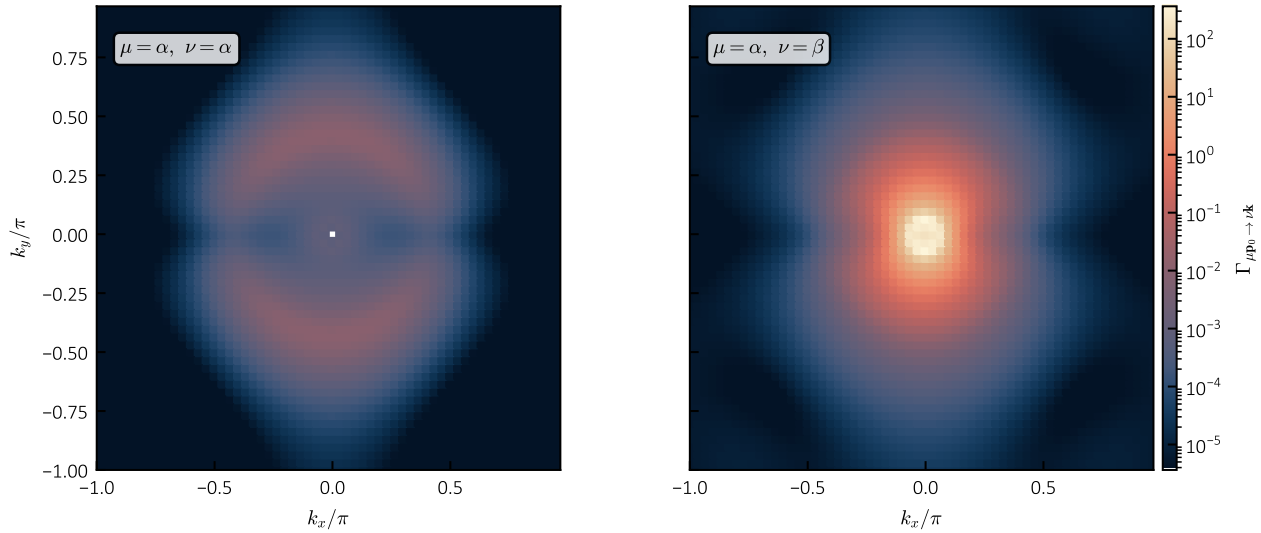
Slika 3 | Verteks pri $\mathbf{d} = (2,0,0)$ v x smeri za $\mathbf{p}_0 = (0,0)$ za obe kopeli z disperzijo: $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$. Dipol zlomi $C_{4v} \to C_{2v}$ simetrijo.



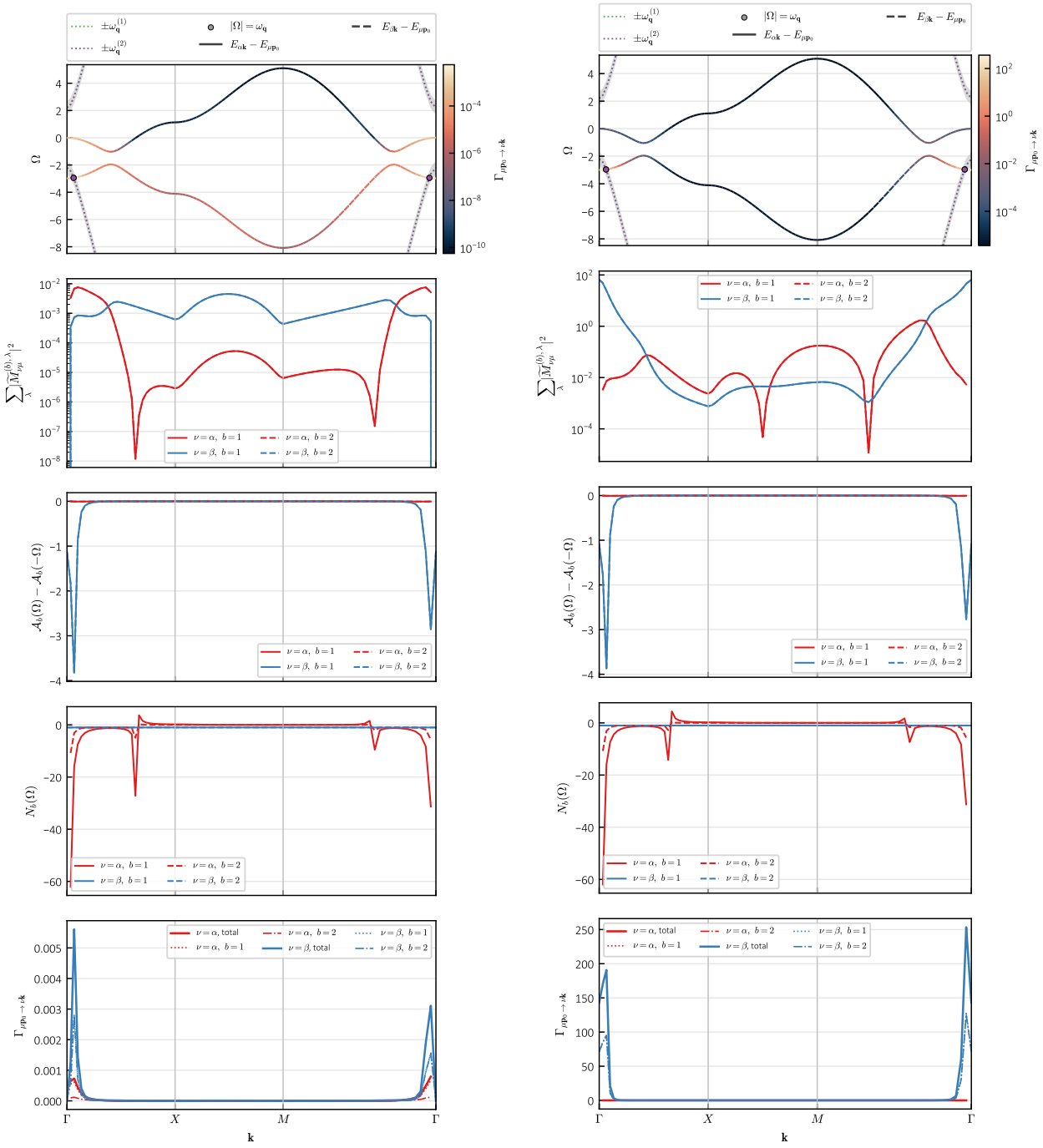
Slika 4 | Komponente faktorjev za kopeli pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$ (majhni q prehodi).



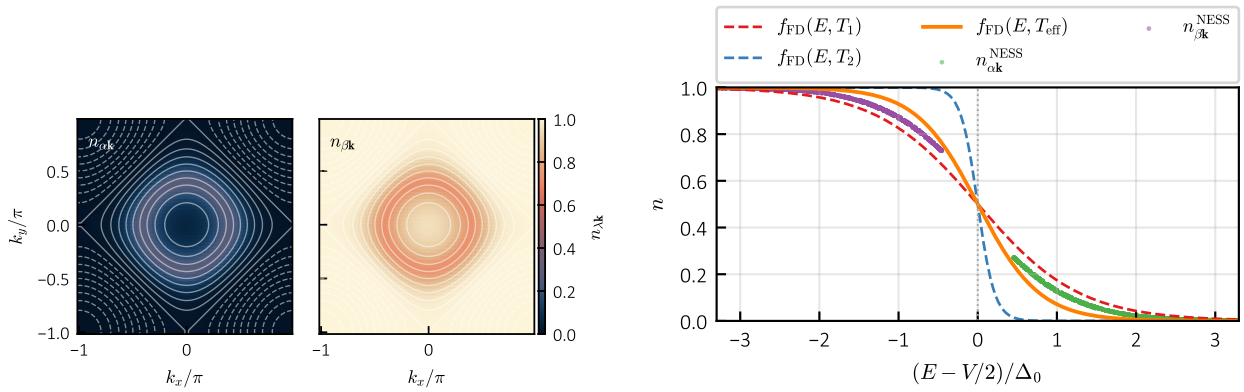
Slika 5 | Končni transfer rate pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$, $\mathbf{d} = (0, 0, 0)$.



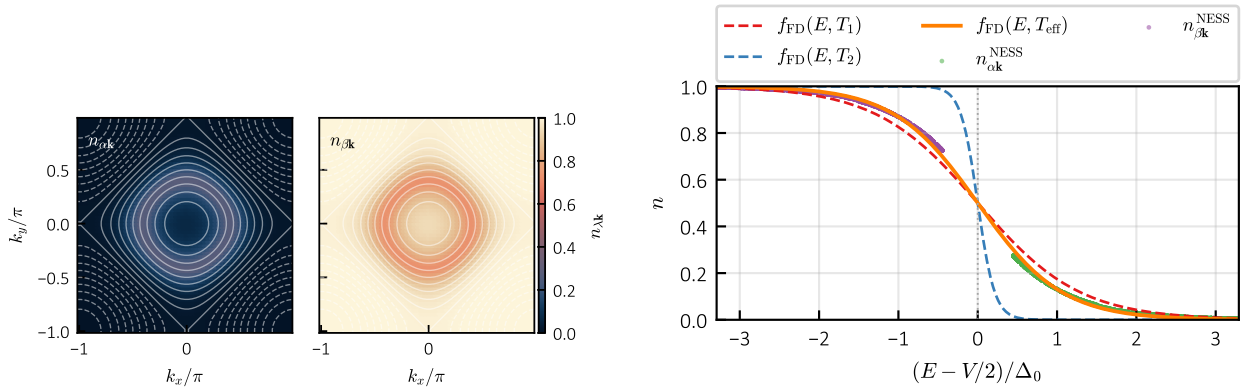
Slika 6 | Končni transfer rate pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$, $\mathbf{d} = 2(1, 0, 0)$.



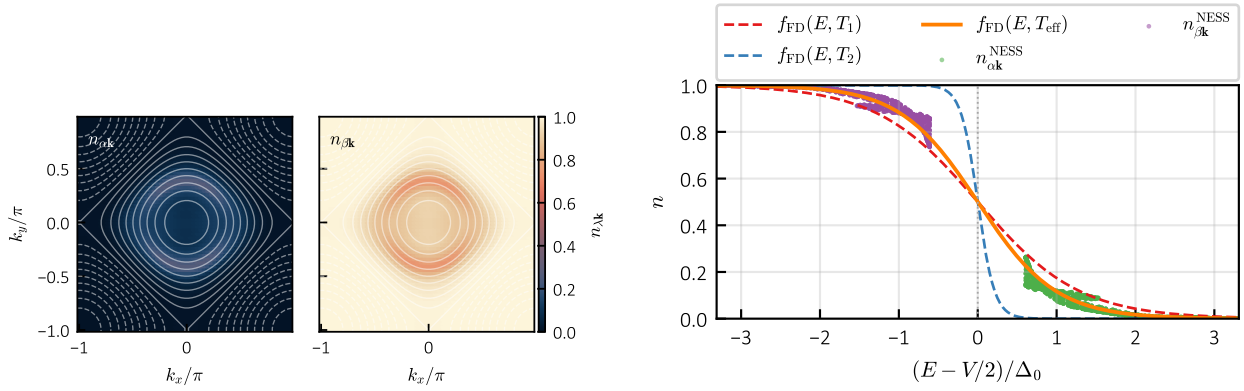
Slika 7 | Komponente faktorjev za kopeli pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$ (majhni q prehodi).



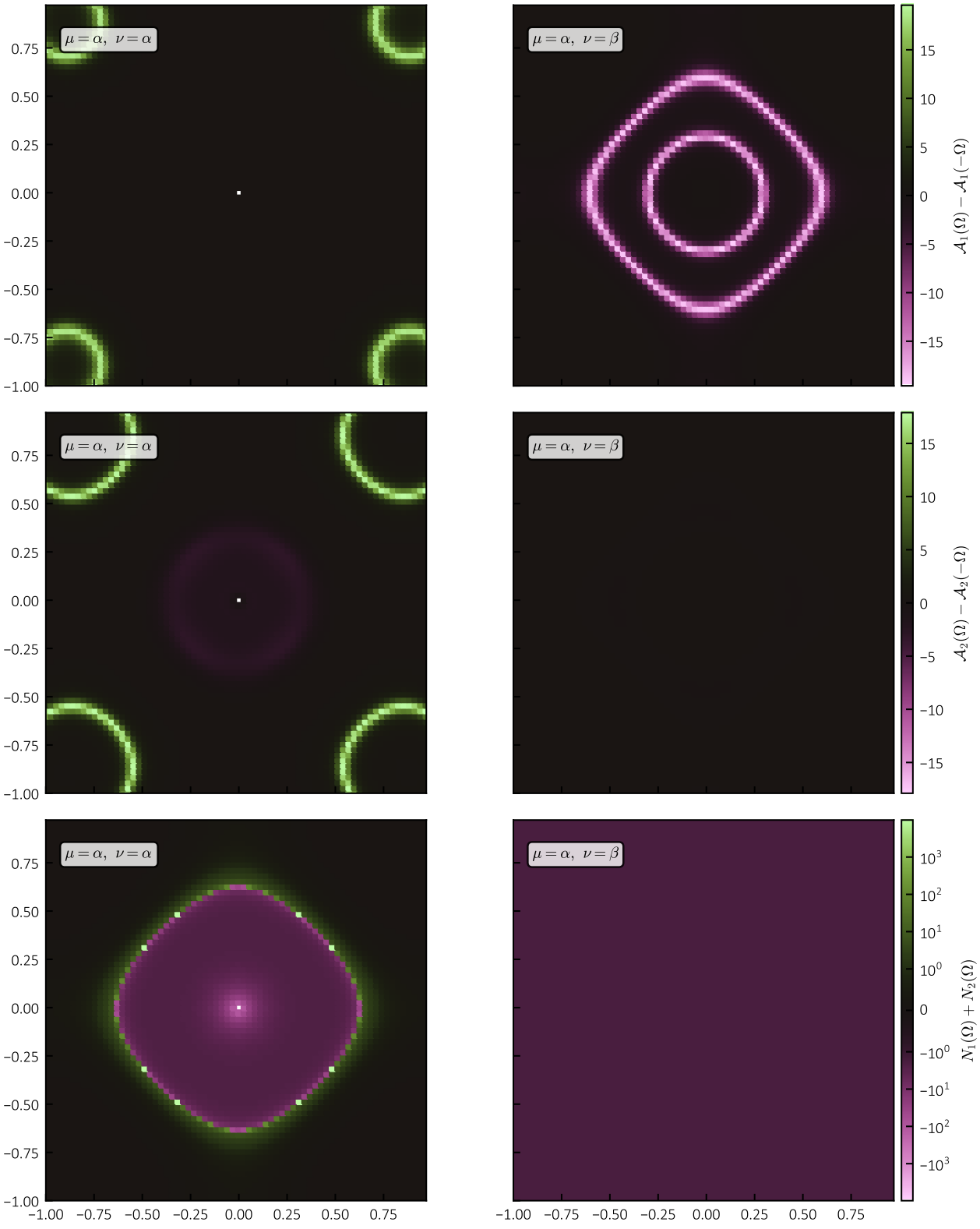
Slika 8 | Komponente faktorjev za kopeli pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$ (majhni q prehodi).



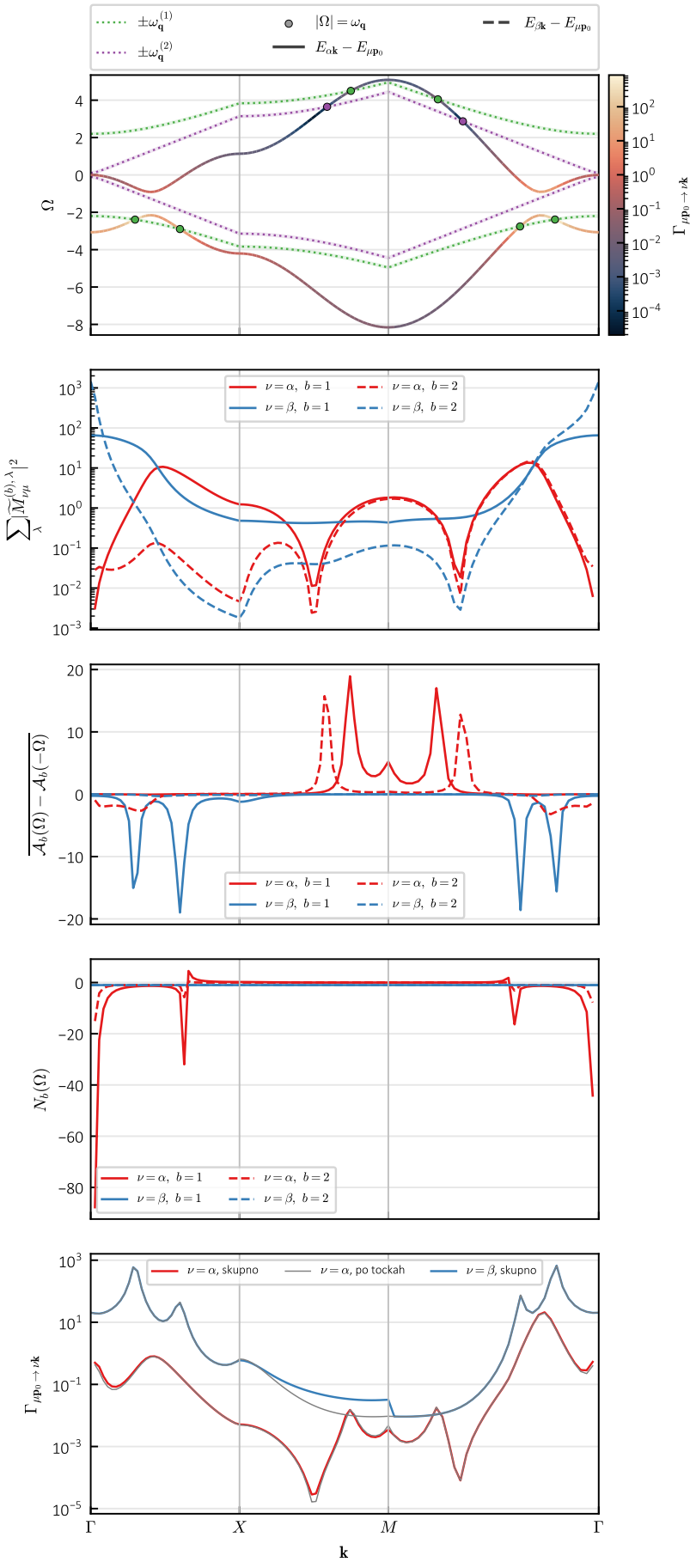
Slika 9 | Komponente faktorjev za kopeli pri $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$, $c = 10, \omega_0 = 2.2$ (majhni q prehodi).



Slika 10 | Komponente faktorjev za kopeli pri mixed disperzijah: $c = 1.0, \omega_0^{(2)} = 2.2, \omega_0^{(1)} = 0.1$ in $\mathbf{d} = 2(1,0,0)$ zdaj zlomi simetrijo v $n(s) \to n(s,u)$, zgleda kar C_{2v}.



Slika 11 | Komponente faktorjev za kopeli pri $c = 1.0, \omega_0^{(1)} = 2.2, \omega_0^{(2)} = 0.1$ in $\mathbf{d} = 2(1, 0, 0)$.



Slika 12 | Mixed disperzije: $c = 1.0, \omega_0^{(1)} = 2.2, \omega_0^{(2)} = 0.1$ in $\mathbf{d} = 2(1, 0, 0)$.

4.1 Numerika detajli (konvergenca in povprecenje)

Za manjše mreže nas zanima kaj se zgodi, če zgrešimo vrh resonance, ker je energija poračunana v neenakomerno razporejenih točkah. Zaradi tega lahko pogledamo, kaj se zgodi s povprečenjem čez sosednje točke. V 1D primeru je mreža definirana v točkah

$$k_i = i\,\delta k, \qquad \mathcal{C}_i = \left[k_i - \tfrac{\delta k}{2}, k_i + \tfrac{\delta k}{2}\right], \qquad q_{ij} = k_i - k_j, \qquad \Omega_{ij} = E_\nu(k_i) - E_\mu(k_j), \tag{244}$$

$$\mathcal{S}(\Omega,\omega) = \mathcal{A}(\Omega) - A(-\Omega) = \frac{8\gamma\,\omega\,\Omega}{\left[(\Omega-\omega)^2+\gamma^2\right]\left[(\Omega+\omega)^2+\gamma^2\right]}\,, \qquad \omega_q = \sqrt{\omega_0^2 + c_s^2 q^2}\,. \tag{245}$$

Po točkah bi računali:

$$\mathcal{S}_{ij}^{\mathrm{pt}} = \mathcal{S}\big(\Omega_{ij}, \omega_{q_{ij}}\big). \tag{246}$$

V svojih celicah sta k in p porazdeljena enakomerno, zato lahko napišemo povprečje čez celico kot

$$\overline{\mathcal{S}}_{ij} = \frac{1}{\delta k^2}\int_{\mathcal{C}_i}\mathrm{d}k\int_{\mathcal{C}_j}\mathrm{d}p\;\mathcal{S}\big(E_\nu(k) - E_\mu(p), \omega_{k-p}\big) = \int\mathrm{d}\Omega\int\mathrm{d}q\;\rho_{ij}(\Omega,q)\,\mathcal{S}(\Omega,\omega_q)\,, \tag{247}$$

z vrjetnostno gostoto:

$$\rho_{ij}(\Omega,q) = \frac{1}{\delta k^2}\int_{\mathcal{C}_i}\mathrm{d}k\int_{\mathcal{C}_j}\mathrm{d}p\;\delta\big(\Omega - E_\nu(k) + E_\mu(p)\big)\,\delta\big(q - k + p\big). \tag{248}$$

Naredil sem približek, kjer razlike energij odčitamo točno v diskretnih točkah, povprečimo pa čez momente (skupna verjentostna gostota za razliko dveh enakomernih spremeljivk $q = k - p$ je konvolucija, torej trikotna funkcija P)

$$\rho_{ij}(\Omega,q) \;\approx\; \delta(\Omega - \Omega_{ij})\,P(q - q_{ij}), \qquad P(x) = \frac{1}{\delta k}\Big(1 - \frac{|x|}{\delta k}\Big)\,\Theta(\delta k - |x|), \qquad \int P = 1\,, \tag{249}$$

kar nam da skupaj pri fiksni energijski razliki $\Omega_{ij} = E_\nu(k_i) - E_\mu(p_j)$:

$$\overline{\mathcal{S}}_{ij} \approx \int_{q_{ij}-\delta k}^{q_{ij}+\delta k} dq\;P(q - q_{ij})\,\mathcal{S}(\Omega_{ij},\omega_q) \tag{250}$$

V praksi to pomeni, da vzamemo n_s točk in seštejemo namest integriramo:

$$t_l = 2\Big[\frac{l-\frac{1}{2}}{n_s} - \frac{1}{2}\Big], \qquad , \qquad \sum_{l=1}^{n_s}\frac{2}{n_s}\big(1 - |t_l|\big) = 1\,(n_s\ \mathrm{sod}), \qquad l = 1,\ldots,n_s, \tag{251}$$

oz. diskretno (z upeljavo $t = (q - q_{ij})/\delta k$) za arbitrarno funkcijo $F(q)$:

$$\int_{-1}^{1}\mathrm{d}t\;(1-|t|)\,F(q_{ij} + t\delta k) = \sum_l \frac{2}{n_s}\big(1 - |t_l|\big)F(q_{ij} + t_l\delta k) + \mathcal{O}(n_s^{-2}). \tag{252}$$

V principu se lahko tu ustavimo, ampak je računanje za vsak par Ω_{ij} zahtevno $\mathcal{O}((2N_k^2)^2 n_s^2)$ (veliko parov $\mathbf{q}_{ij}$ da enako energijo, ta pa se z iteracijami spreminja, zato ga tabeliram in v vsaki iteraciji odčitam). Zato tabeliramo razlike energij in interpoliramo. Izbral sem $\delta\Omega = \gamma/8$, tako je $N_\Omega = \lceil E_{\max}/\delta\Omega\rceil + 2$, kjer si $E_{\max}$ izberemo tako, da je večji od največje možne razlike energij[1]. Za neko funkcijo $F(\Omega)$ potem računamo le N_Ω točk in za določen $\Omega_n \le \Omega_{ij} \le \Omega_{n+1}$ interpoliramo linearno:

$$F(\Omega) = (1-\alpha)F(\Omega_n) + \alpha F(\Omega_{n+1})\,, \quad \alpha = |\Omega|/\delta\Omega - \lfloor|\Omega|/\delta\Omega\rfloor \tag{253}$$

to skupaj stane $N_k^2 N_\Omega n_s^2 + (2N_k)^2$, dodatni člen je za odčitavanje energije. Na koncu se bolj splača skupaj povprečiti ne le spektralno funkcijo, temveč tudi dodatne faktorje. S krajšo notacijo:

$$\mathbf{h} = \Big(\sin\big(k'_x + \tfrac{q_x}{2}\big),\ \sin\big(k'_y + \tfrac{q_y}{2}\big)\Big) \tag{254}$$

in

$$Z_{xx} = \sin^2\varphi + (\omega_0/\omega_{\mathbf{q}})^2\cos^2\varphi, \qquad Z_{yy} = \cos^2\varphi + (\omega_0/\omega_{\mathbf{q}})^2\sin^2\varphi, \qquad Z_{xy} = 2\sin\varphi\cos\varphi\,((\omega_0/\omega_{\mathbf{q}})^2 - 1), \tag{255}$$

[1] V kodi to ocenim avtomatsko navzgor z 10% maržo.

definiramo še (bolj stabilno v kodi, tudi produkt $\Omega N(\Omega)$ je stabilen za $\Omega \to 0$):

$$F_{\mathbf{q}}(\Omega) = \frac{\mathcal{A}_{\mathbf{q}}(\Omega) - \mathcal{A}_{\mathbf{q}}(-\Omega)}{\Omega\omega_{\mathbf{q}}} = \frac{8\gamma}{\left[(\Omega-\omega)^2+\gamma^2\right]\left[(\Omega+\omega)^2+\gamma^2\right]}\,, \tag{256}$$

tako je po točkah (to je enaka matrika kot (243))

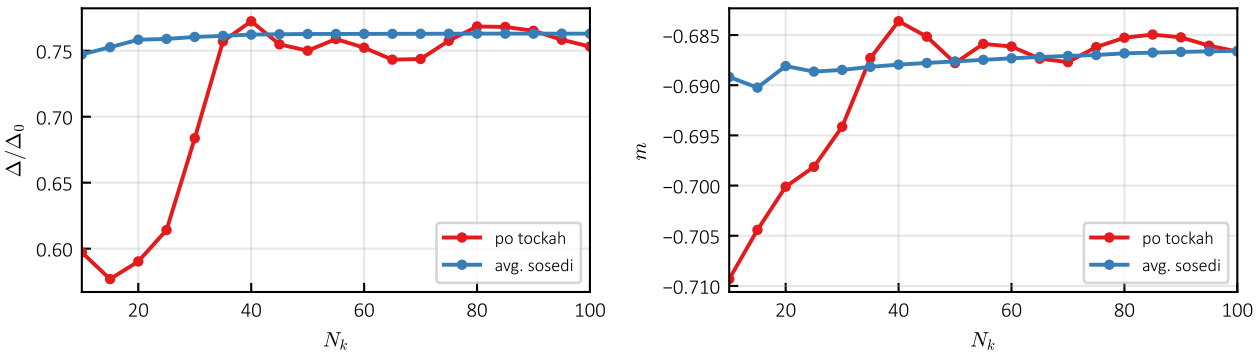
$$\Gamma_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} = A\cdot\Omega N(\Omega)\cdot\left[\left(h_x^2 Z_{xx}F_{\mathbf{q}} + h_y^2 Z_{yy}F_{\mathbf{q}} + h_x h_y Z_{xy}F_{\mathbf{q}}\right)C^2_{\gamma\gamma'} + Z_{xx}F_{\mathbf{q}}D^2_{\gamma\gamma'}\right], \tag{257}$$

Povprečimo produkte tipa $Z_{xx}F_{\mathbf{q}}$ s trikotno funkcijo, medtem ko $\mathbf{h}$, $C_{\gamma,\gamma'}$ in Ω računamo le v centrih celic. Povprečenje $N(\Omega)$ po energiji bi lahko zlomilo detailed balance glede na mrežo. Faktorji $\mathbf{h}$ v argumentu ne vsebujejo razlike momentov (če je $\mathbf{k} = \mathbf{k}_i + \mathbf{x}$ in $\mathbf{k}' = \mathbf{k}'_j + \mathbf{y}$ kjer sta $\mathbf{x}$ in $\mathbf{y}$ naključni spremenljivki v celici, potem povprečimo po $\mathbf{x}-\mathbf{y}$, argument $\mathbf{h}$ pa je $(\mathbf{k}_i + \mathbf{k}'_j)/2 + (\mathbf{x}+\mathbf{y})/2$)

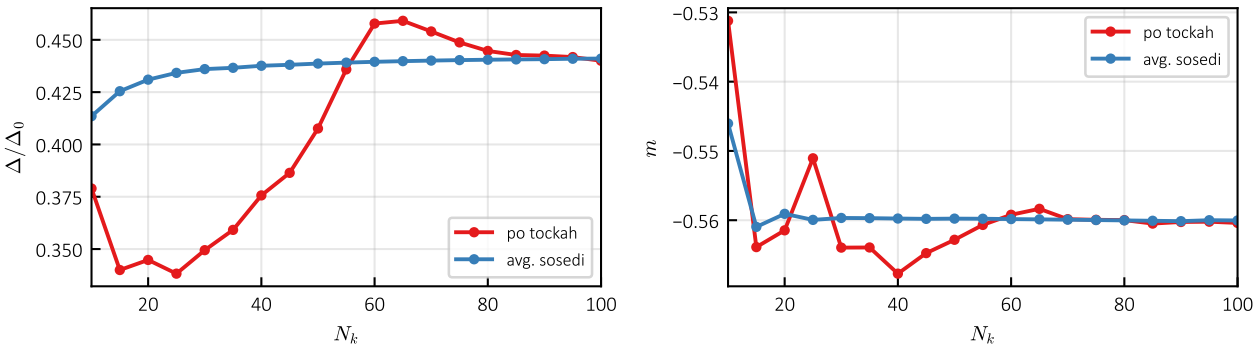
$$\overline{\Gamma}_{\gamma'\mathbf{k}'\to\gamma\mathbf{k}} = A\cdot\Omega N(\Omega)\cdot\left[\left(h_x^2\overline{Z_{xx}F_{\mathbf{q}}} + h_y^2\overline{Z_{yy}F_{\mathbf{q}}} + h_x h_y\overline{Z_{xy}F_{\mathbf{q}}}\right)C^2_{\gamma\gamma'} + \overline{Z_{xx}F_{\mathbf{q}}}D^2_{\gamma\gamma'}\right], \tag{258}$$

kjer je

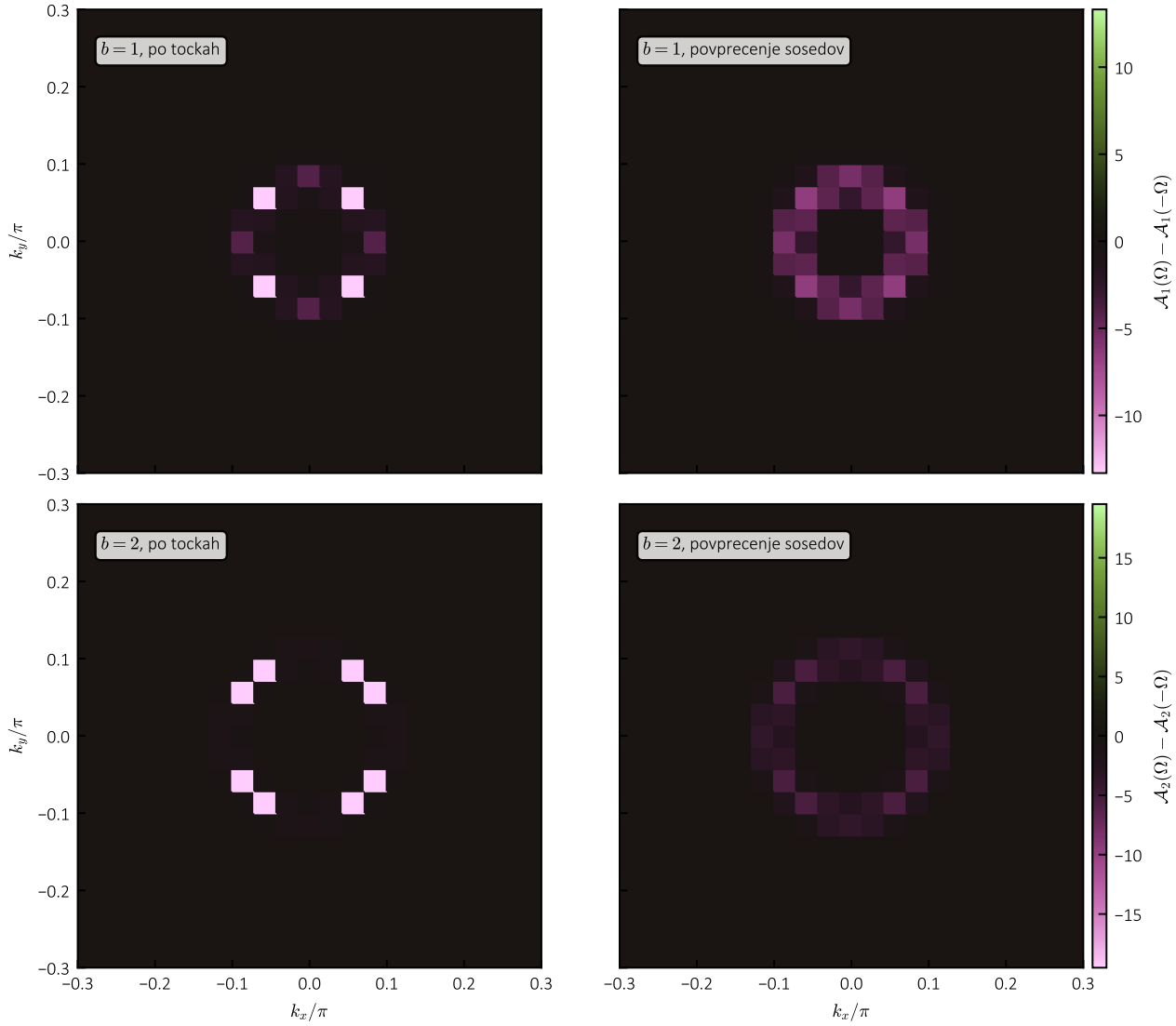
$$\overline{X_{\mathbf{q}}(\Omega)} = \int \mathrm{d}x\,\mathrm{d}y\,P(x)P(y)\,X_{(q_x+x,q_y+y)} \simeq \sum_{l,m}^{n_s}\left(\frac{2}{n_s}\right)^2\left(1-|t_l|\right)\left(1-|t_m|\right)X_{\mathbf{q}+(t_l\delta k_x,\,t_m\delta k_y)}\,. \tag{259}$$



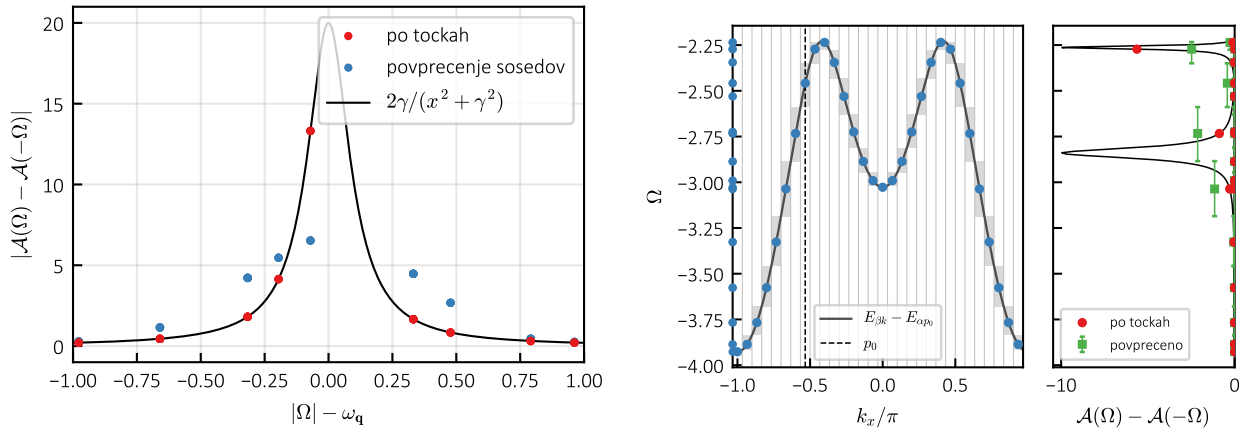
Slika 13 | Mixed disperzije: $c = 10.0$, $\omega_0^{(1)} = 2.2$, $\omega_0^{(2)} = 0.1$ in $\mathbf{d} = 2(1,0,0)$ ter $\gamma = 0.1$. KOnvergenca pri $T_1 = 1.2T_c$ in $T_2 = 0.2T_c$. TO je za primer ko je $q \ll k_F$, tako da so small momentum prehodi, vecinoma interband (diagonalna baza).



Slika 14 | Mixed disperzije: $c = 10.0$, $\omega_0^{(1)} = 2.2$, $\omega_0^{(2)} = 0.1$ in $\mathbf{d} = 2(1,0,0)$ ter $\gamma = 0.1$. KOnvergenca pri $T_1 = 2.5T_c$ in $T_2 = 0.2T_c$. TO je za primer ko je $q \ll k_F$, tako da so small momentum prehodi, vecinoma interband (diagonalna baza) (dlje od diagonale $T_1 = T_2$ pocasneje konvergira).



Slika 15 | .



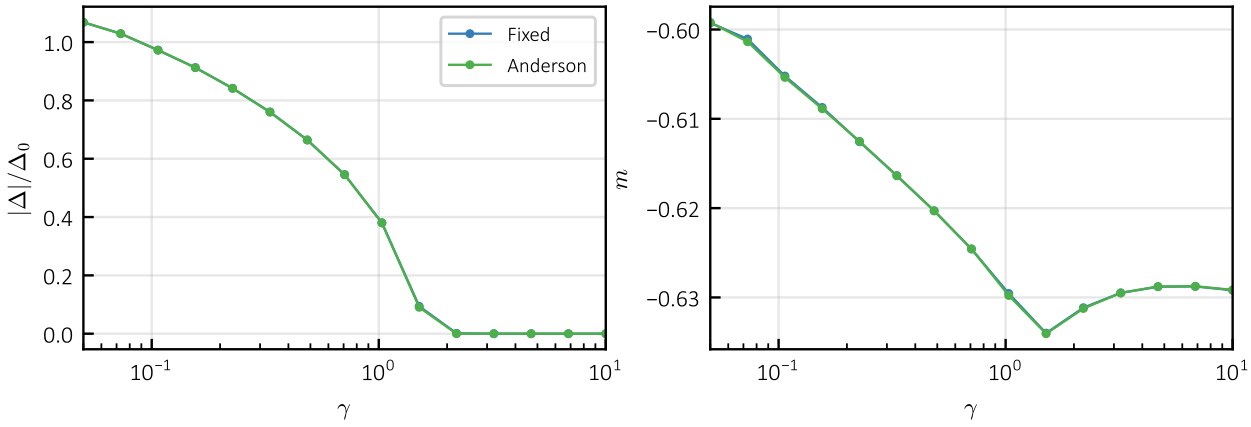
Slika 16 | .

4.1.1 Andersonov algoritem

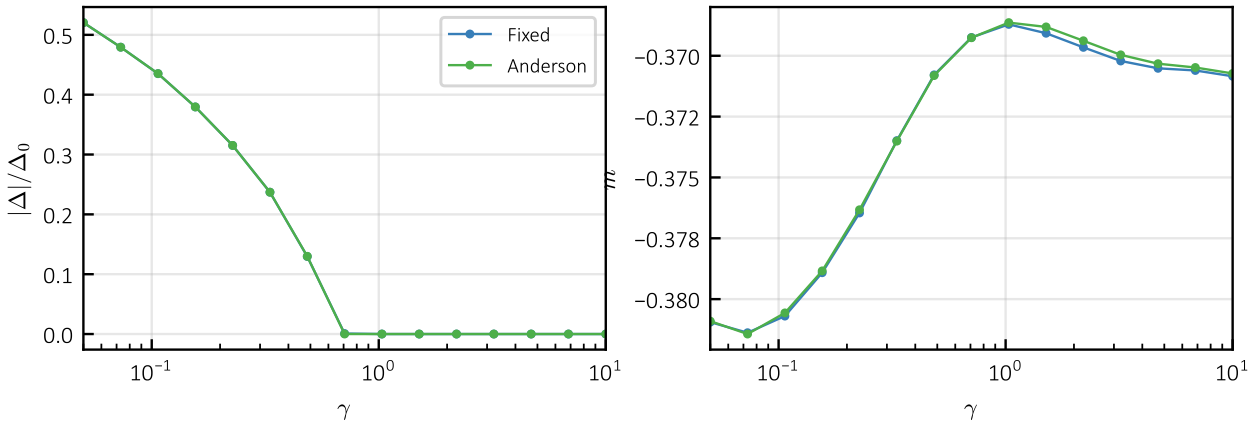
Samokonsistenten sistem enačb je ekvivalenten reševanju fixed point problema $\mathbf{x} = \mathcal{G}(\mathbf{x})$, oziroma iskanje ničel funckije $\mathbf{f}(\mathbf{x}) = \mathcal{G}(\mathbf{x}) - \mathbf{x}$. Pri Picard iteracijah samo posodabljamo s prejšnjo vrednostjo:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \beta\, \mathbf{f}_k \,. \tag{260}$$

okoli rešitve se konvergenca ustavi, saj je: $\mathbf{f}(\mathbf{x}) \approx J(\mathbf{x} - \mathbf{x}^*)$ kjer je Jakobijan $J = \partial\mathcal{G}/\partial\mathbf{x} - \mathbb{1}$.



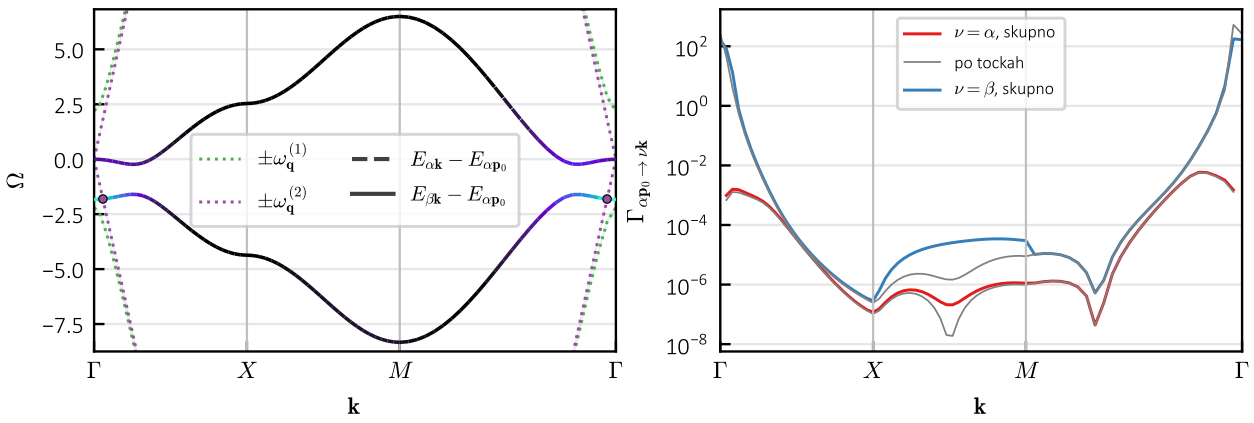
Slika 17 | $G = 2, t_a = -t_b = 1$ in $V = 7.8$, kjer je $\Delta_0 = 1.0$. Disperziji sta: $c = 10.0$, $\omega_0^{(1)} = 3.0$, $\omega_0^{(2)} = 0.01$. V sklopitvi je $d = 2$ in $N_k = 65$, $n_s = 8$. V točki $T_1 = 4.4T_c$ in $T_2 = 0.01T_c$.



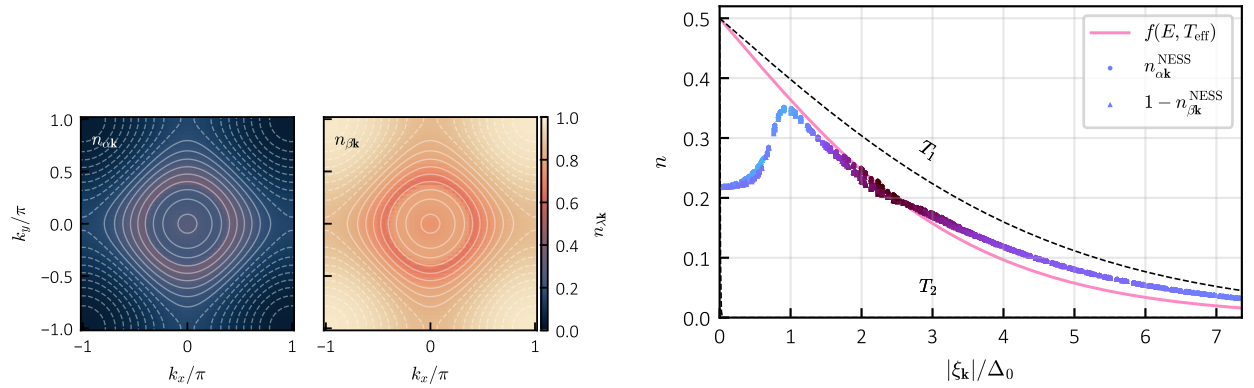
Slika 18 | $G = 2, t_a = -t_b = 1$ in $V = 4.4$, kjer je $\Delta_0 = 1.0$. Disperziji sta: $c = 10.0$, $\omega_0^{(1)} = 3.0$, $\omega_0^{(2)} = 0.01$. V sklopitvi je $d = 2$ in $N_k = 65$, $n_s = 8$. V točki $T_1 = 4.4T_c$ in $T_2 = 0.01T_c$.

4.2 Primer NESS za BEC

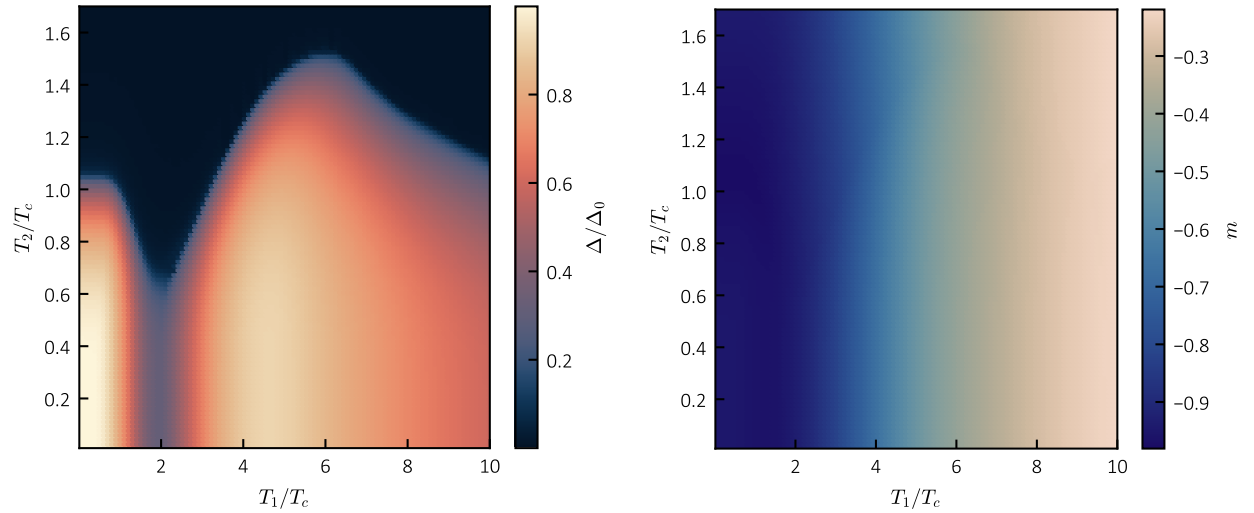
Primer NESS v BEC režimu: $G = 2, t_a = -t_b = 1$ in $V = 7.8$, kjer je $\Delta_0 = 1.0$. Disperziji sta: $c = 10.0$, $\omega_0^{(1)} = 2.2$, $\omega_0^{(2)} = 0.01$. V sklopitvi je $d = 2$ in $\gamma = 0.15$ in $N_k = 50$.



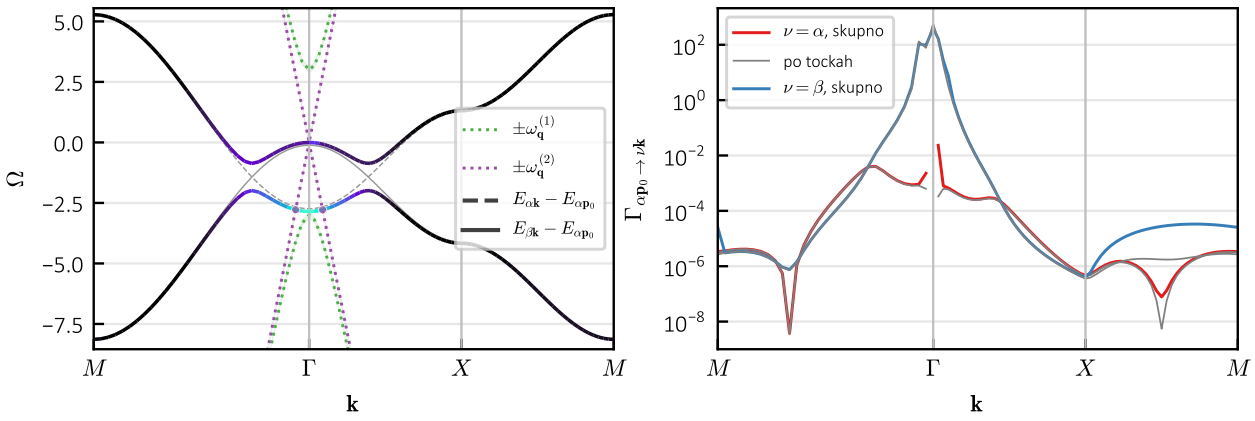
Slika 19 | Grafi so v točki $T_1 = 4.4T_c$ in $T_2 = 0.01T_c$, kjer preidemo v BCS režim.



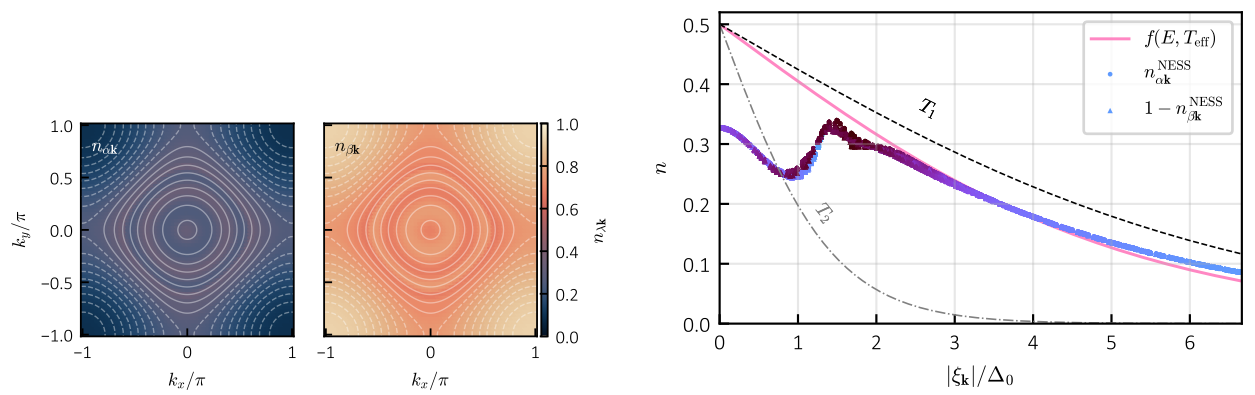
Slika 20 | Grafi so v točki $T_1 = 4.4T_c$ in $T_2 = 0.01T_c$, kjer preidemo v BCS režim.



Slika 21 | Disperziji sta: $c = 10.0, \omega_0^{(1)} = 3, \omega_0^{(2)} = 0.01$, BEC režim.



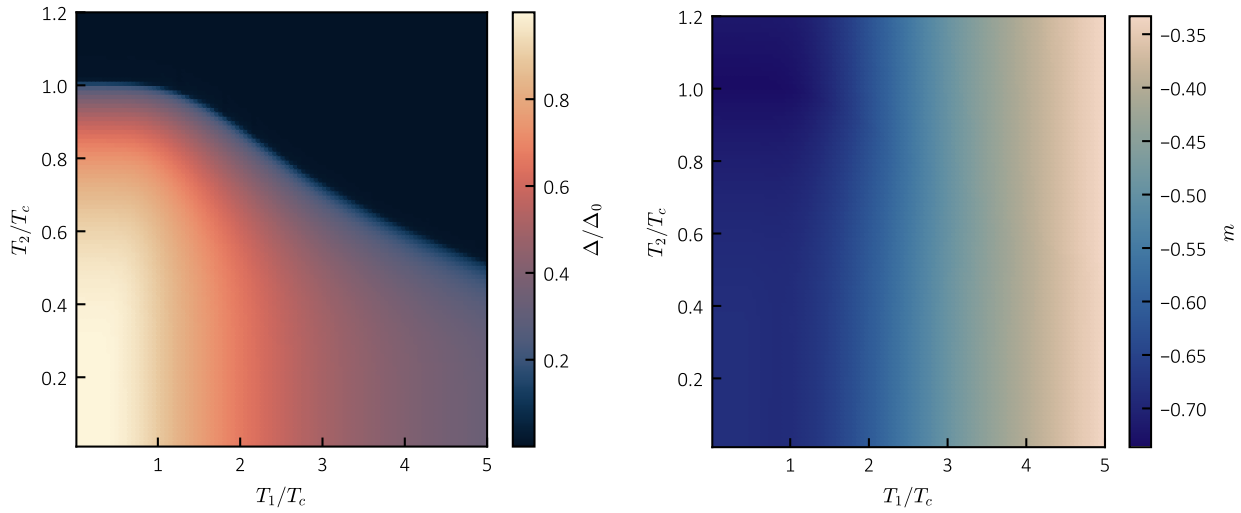
Slika 22 | Disperziji sta: $c = 10.0, \omega_0^{(1)} = 3, \omega_0^{(2)} = 0.01$. Grafi so v točki $T_1 = 6.0T_c$ in $T_2 = 1.3T_c$, kjer preidemo v BCS režim.



Slika 23 | Disperziji sta: $c = 10.0, \omega_0^{(1)} = 3, \omega_0^{(2)} = 0.01$. Grafi so v točki $T_1 = 6.0T_c$ in $T_2 = 1.3T_c$, kjer preidemo v BCS režim.

4.3 Primer NESS za BCS

Primer NESS v BCS režimu: $G = 2, t_a = -t_b = 1$ in $V = 4.4$, kjer je $\Delta_0 = 1.0$. Disperziji sta: $c = 10.0$, $\omega_0^{(1)} = 3.0$, $\omega_0^{(2)} = 0.01$. V sklopitvi je $d = 2$ in $\gamma = 0.15$ in $N_k = 50$.



Slika 24 | Disperziji sta: $c = 10.0$, $\omega_0^{(1)} = 3$, $\omega_0^{(2)} = 0.01$, BCS režim. Obe s fotonskim verteksom.

5 Pretvorbe enot

Do zdaj smo uporabljali $\hbar = k_B = 1$ in delali v enotah $a_0 = 1$ ter $t_a = 1$. Za pretvorbe si moramo izbrati konkretne vrednosti za $a_0 = 10$ Å in $t_a = 1\,\mathrm{eV}$. Za brezdimenzijske količine bom uporabljal kačo, le v tem odseku. Od tod so naravne pretvorbe za hitrosti:

$$v_0 = \frac{t_a a_0}{\hbar} \simeq 1519\,\mathrm{km/s} \tag{261}$$

za izbrane vrednosti parametrov je potem hitrost svetlobe $\tilde{c} = c/v_0 \simeq 200$. Za hitrost zvoka $\tilde{c}_s = 5\,\mathrm{km/s}/v_0 \simeq 0.0033$. Pretvorba za količine povezane s časom je

$$\tau_0 = \frac{\hbar}{t_a} \simeq 0.66\,\mathrm{fs} \tag{262}$$

in torej $\tilde{\omega}_0 = \tau_0 \omega_0 \simeq 0.66\,\mathrm{fs} \cdot 1.5 \cdot 10^{15}\,\mathrm{Hz} = 1$. Dolžina resonatorja je v tem primeru $\tilde{L} = L/a_0 = \pi\tilde{c}/\tilde{\omega}_0 \simeq 620$. Za fotonsko sklopitveno konstanto velja:

$$\tilde{g} = \frac{g}{t_a^2} = \frac{e^2\hbar}{2\mathcal{V}\varepsilon_0\omega_{\mathbf{q}}}\left(\frac{t_a a_0}{\hbar}\right)^2 \frac{1}{t_a^2} = 2\pi\alpha\frac{\tilde{c}}{\tilde{\omega}_{\mathbf{q}}}\frac{a_0^3}{\mathcal{V}} = 2\alpha\frac{\tilde{\omega}_0}{\tilde{\omega}_{\mathbf{q}}}\frac{1}{N_k^2}\,. \tag{263}$$

Tu sem za volumen vzel $\mathcal{V} = N_k^2 a_0^3 \tilde{L}$. Uteži $1/N_k^2$ so že vštete v vsote in povprečja po momentih, zato je v končnem izrazu ne upoštevamo. V kodi ja tako vrednost za amplitudo

$$\tilde{A} = \tilde{g}\tilde{\omega}_{\mathbf{q}} = 2\alpha\tilde{\omega}_0 = \frac{2}{137}\tilde{\omega}_0\,. \tag{264}$$

Drugi red je sorazmeren z $\tilde{A}^2 \ll \tilde{A}$. Za fonone velja

$$\tilde{g} = \frac{g}{t_a^2} = \frac{\hbar}{2MN_k^2\omega_{\mathbf{q}}} = \frac{1}{2\tilde{M}}\frac{1}{N_k^2}\frac{1}{\tilde{\omega}_{\mathbf{q}}} \tag{265}$$

kjer je brezdimenzijska masa ionov $\tilde{M} = M\frac{a_0^2 t_a}{\hbar^2} \simeq 23900M[\mathrm{u}]$. V kodi je torej amplituda za fononsko sklopitev

$$\tilde{A} = \frac{1}{2\tilde{M}} \tag{266}$$

razmerje med amplitudama je

$$\frac{\tilde{g}_{\text{fon.}}}{\tilde{g}_{\text{fot.}}} = \frac{\sqrt{\tilde{\omega}_0^2 + q^2\tilde{c}^2}}{4\alpha\tilde{M}\tilde{\omega}_0|q|\tilde{c}_s}\,, \tag{267}$$

torej za $q \to 0$ fononi dominirajo v verteksu:

$$\frac{\tilde{g}_{\text{fon.}}}{\tilde{g}_{\text{fot.}}} = \frac{1}{q}\frac{1}{4\alpha\tilde{M}\tilde{c}_s} \tag{268}$$

za večje momente pa:

$$\frac{\tilde{g}_{\text{fon.}}}{\tilde{g}_{\text{fot.}}} = \frac{\tilde{c}}{\tilde{c}_s}\frac{1}{4\alpha\tilde{M}\tilde{\omega}_0} \tag{269}$$

realistični dipol znotraj celice znaša okoli $d = 0.1 - 1$ Å, torej $\tilde{d} \simeq 0.01 - 0.1$.

Literatura

[1] Monique Combescot in Shiue-Yuan Shiau. *Excitons and cooper pairs: two composite bosons in many-body physics*. Oxford graduate texts. Oxford: Oxford university press, 2016. ISBN: 978-0-19-875373-5.

[2] D. Jérome, T. M. Rice in W. Kohn. "Excitonic Insulator". V: *Physical Review* 158.2 (10. jun. 1967), str. 462–475. ISSN: 0031-899X. DOI: 10.1103/PhysRev.158.462. URL: https://link.aps.org/doi/10.1103/PhysRev.158.462 (pridobljeno 27. 6. 2026).

[3] A. Asenjo-Garcia in sod. "Exponential Improvement in Photon Storage Fidelities Using Subradiance and "Selective Radiance" in Atomic Arrays". V: *Physical Review X* 7.3 (3. avg. 2017), str. 031024. ISSN: 2160-3308. DOI: 10.1103/PhysRevX.7.031024. URL: http://link.aps.org/doi/10.1103/PhysRevX.7.031024 (pridobljeno 27. 6. 2026).

[4] Gianluca Lagnese in sod. *Neural network modeling of many-body super- and sub-radiant dynamics*. 6. maj 2026. DOI: 10.48550/arXiv.2605.04640. arXiv: 2605.04640[quant-ph]. URL: http://arxiv.org/abs/2605.04640 (pridobljeno 27. 6. 2026).

[5] Kwangrae Kim in sod. "Direct observation of excitonic instability in Ta2NiSe5". V: *Nature Communications* 12.1 (30. mar. 2021), str. 1969. ISSN: 2041-1723. DOI: 10.1038/s41467-021-22133-z. arXiv: 2007.08212[cond-mat.str-el]. URL: http://arxiv.org/abs/2007.08212 (pridobljeno 27. 6. 2026).

[6] Yuta Murakami in sod. "Collective Modes in Excitonic Insulators: Effects of Electron-Phonon Coupling and Signatures in Optical Response". V: *Physical Review B* 101.19 (11. maj 2020), str. 195118. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/PhysRevB.101.195118. arXiv: 2003.10789[cond-mat.str-el]. URL: http://arxiv.org/abs/2003.10789 (pridobljeno 28. 6. 2026).

[7] Tatsuya Kaneko in Yukinori Ohta. "A New Era of Excitonic Insulators". V: *Journal of the Physical Society of Japan* 94.1 (15. jan. 2025), str. 012001. ISSN: 0031-9015, 1347-4073. DOI: 10.7566/JPSJ.94.012001. arXiv: 2411.10985[cond-mat.str-el]. URL: http://arxiv.org/abs/2411.10985 (pridobljeno 28. 6. 2026).

[8] Daniel Arovas. "Lecture Notes on Condensed Matter Physics (A Work in Progress)". V: ().

[9] B. Hülsen in sod. "Phase diagram of the excitonic insulator". V: *Physica B: Condensed Matter* 378-380 (maj 2006), str. 267–268. ISSN: 09214526. DOI: 10.1016/j.physb.2006.01.099. URL: https://linkinghub.elsevier.com/retrieve/pii/S092145260600130X (pridobljeno 9. 8. 2026).

[10] V. Apinyan in T. K. Kopeć. "Phase coherence and spectral functions in the two-dimensional excitonic systems". V: *Physica B: Condensed Matter* 473 (sep. 2015), str. 75–92. ISSN: 09214526. DOI: 10.1016/j.physb.2015.05.027. arXiv: 1311.1645[cond-mat.str-el]. URL: http://arxiv.org/abs/1311.1645 (pridobljeno 9. 8. 2026).

[11] D. I. Golosov. "Excitonic Insulator and the Extended Falicov–Kimball Model Away from Half-Filling". V: *Physical Review B* 112.24 (26. dec. 2025), str. 245161. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/l9zh-ldpv. arXiv: 2507.13967[cond-mat.str-el]. URL: http://arxiv.org/abs/2507.13967 (pridobljeno 9. 8. 2026).

[12] Farkašovský. "DMRG study of exciton condensation in the extended Falicov-Kimball model". V: *Condensed Matter Physics* 23.4 (dec. 2020), str. 43709. ISSN: 1607324X, 22249079. DOI: 10.5488/CMP.23.43709. URL: http://www.icmp.lviv.ua/journal/zbirnyk.104/43709/abstract.html (pridobljeno 9. 8. 2026).

[13] Francesco Campaioli, Jared H. Cole in Harini Hapuarachchi. "Quantum Master Equations: Tips and Tricks for Quantum Optics, Quantum Computing, and Beyond". V: *PRX Quantum* 5.2 (10. jun. 2024), str. 020202. ISSN: 2691-3399. DOI: 10.1103/PRXQuantum.5.020202. URL: https://link.aps.org/doi/10.1103/PRXQuantum.5.020202 (pridobljeno 12. 8. 2026).

[14] Iris Ulčakar in Zala Lenarčič. "Generalized Gibbs ensembles in weakly interacting dissipative systems and digital quantum computers". V: *SciPost Physics* 19.3 (4. sep. 2025), str. 068. ISSN: 2542-4653. DOI: 10.21468/SciPostPhys.19.3.068. URL: https://scipost.org/10.21468/SciPostPhys.19.3.068 (pridobljeno 13. 8. 2026).

[15] E. Ilievski in sod. "Complete Generalized Gibbs Ensembles in an Interacting Theory". V: *Physical Review Letters* (2015). DOI: 10.1103/PhysRevLett.115.157201. URL: https://sci-hub.box/10.1103/PhysRevLett.115.157201 (pridobljeno 13. 8. 2026).

[16] Marcos Rigol in sod. "Relaxation in a Completely Integrable Many-Body Quantum System: An Ab Initio Study of the Dynamics of the Highly Excited States of Lattice Hard-Core Bosons". V: *Physical Review Letters* 98.5 (1. feb. 2007), str. 050405. ISSN: 0031-9007, 1079-7114. DOI: 10.1103/PhysRevLett.98.050405. arXiv: cond-mat/0604476. URL: http://arxiv.org/abs/cond-mat/0604476 (pridobljeno 13. 8. 2026).

[17] Scott Davidson in sod. "The dark side of energy transport along excitonic wires: On-site energy barriers facilitate efficient, vibrationally mediated transport through optically dark subspaces". V: *The Journal of Chemical Physics* (2020). DOI: 10.1063/5.0023702. URL: https://sci-hub.box/10.1063/5.0023702 (pridobljeno 16. 8. 2026).

[18] Florian Lange, Zala Lenarčič in Achim Rosch. "Pumping approximately integrable systems". V: *Nature Communications* 8.1 (9. jun. 2017), str. 15767. ISSN: 2041-1723. DOI: 10.1038/ncomms15767. arXiv: 1608.03563[cond-mat.str-el]. URL: http://arxiv.org/abs/1608.03563 (pridobljeno 16. 8. 2026).

[19] Cristian Zanoci, Yongchan Yoo in Brian Swingle. "Thermalization at Low Temperatures via Weakly-Damped Multi-Site Baths". V: *Physical Review B* 108.3 (31. jul. 2023), str. 035156. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/PhysRevB.108.035156. arXiv: 2304.08525[cond-mat.str-el]. URL: http://arxiv.org/abs/2304.08525 (pridobljeno 17. 8. 2026).

[20] Ana Štuhec. "Termoelektrični pojav v ekscitonskih izolatorjih". V: ().

[21] Andrew J. Daley. "Quantum trajectories and open many-body quantum systems". V: *Advances in Physics* 63.2 (4. mar. 2014), str. 77–149. ISSN: 0001-8732, 1460-6976. DOI: 10.1080/00018732.2014.933502. URL: http://www.tandfonline.com/doi/abs/10.1080/00018732.2014.933502 (pridobljeno 19. 8. 2026).

[22] Elahe Davari in Mehdi Kargarian. "Probing the collective excitations of excitonic insulators in an optical cavity". V: *Physical Review B* 112.16 (17. okt. 2025), str. 165124. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/zqhl-kncx. arXiv: 2501.04736[cond-mat.mes-hall]. URL: http://arxiv.org/abs/2501.04736 (pridobljeno 25. 8. 2026).

[23] E. T. Jaynes in F. W. Cummings. "Comparison of quantum and semiclassical radiation theories with application to the beam maser". V: *Proceedings of the IEEE* (1963). DOI: 10.1109/PROC.1963.1664. URL: https://sci-hub.box/10.1109/PROC.1963.1664 (pridobljeno 25. 8. 2026).

[24] Anneswa Paul in Upendra Harbola. *Light-matter interaction inside an optical cavity: A perspective*. 27. apr. 2025. DOI: 10.48550/arXiv.2504.14266. arXiv: 2504.14266[quant-ph]. URL: http://arxiv.org/abs/2504.14266 (pridobljeno 25. 8. 2026).

[25] Arkajit Mandal in sod. *Theoretical Advances in Polariton Chemistry and Molecular Cavity Quantum Electrodynamics*. 7. dec. 2022. DOI: 10.26434/chemrxiv-2022-g9lr7. URL: https://chemrxiv.org/doi/full/10.26434/chemrxiv-2022-g9lr7 (pridobljeno 25. 8. 2026).

[26] Jiajun Li in sod. "Electromagnetic coupling in tight-binding models for strongly correlated light and matter". V: *Physical Review B* 101.20 (21. maj 2020), str. 205140. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/PhysRevB.101.205140. arXiv: 2001.09726[cond-mat.str-el]. URL: http://arxiv.org/abs/2001.09726 (pridobljeno 25. 8. 2026).

[27] Simon B. Jäger in sod. "Lindblad master equations for quantum systems coupled to dissipative bosonic modes". V: *Physical Review Letters* 129.6 (2. avg. 2022), str. 063601. ISSN: 0031-9007, 1079-7114. DOI: 10.1103/PhysRevLett.129.063601. arXiv: 2203.03302[quant-ph]. URL: http://arxiv.org/abs/2203.03302 (pridobljeno 25. 8. 2026).

[28] Gerald Dunne in Roman Jackiw. ""Peireles Substitution" and Chern-Simons Quantum Mechanics". V: *Nuclear Physics B - Proceedings Supplements* 33.3 (nov. 1993), str. 114–118. ISSN: 09205632. DOI: 10.1016/0920-5632(93)90376-H. arXiv: hep-th/9204057. URL: http://arxiv.org/abs/hep-th/9204057 (pridobljeno 26. 8. 2026).

[29] Florian Lange, Zala Lenarčič in Achim Rosch. "Time-dependent generalized Gibbs ensembles in open quantum systems". V: *Physical Review B* 97.16 (26. apr. 2018), str. 165138. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/PhysRevB.97.165138. arXiv: 1801.07646[cond-mat.str-el]. URL: http://arxiv.org/abs/1801.07646 (pridobljeno 2. 9. 2026).

[30] Jonathan B. Curtis in sod. "Cavity Quantum Eliashberg Enhancement of Superconductivity". V: *Physical Review Letters* 122.16 (26. apr. 2019), str. 167002. ISSN: 0031-9007, 1079-7114. DOI: 10.1103/PhysRevLett.122.167002. URL: https://link.aps.org/doi/10.1103/PhysRevLett.122.167002 (pridobljeno 10. 9. 2026).

[31] Mona H. Kalthoff in sod. "Nonequilibrium phase transition in a driven-dissipative quantum antiferromagnet". V: *Physical Review Research* 4.2 (12. maj 2022), str. 023115. ISSN: 2643-1564. DOI: 10.1103/PhysRevResearch.4.023115. URL: https://link.aps.org/doi/10.1103/PhysRevResearch.4.023115 (pridobljeno 16. 9. 2026).